

A transport model and numerical simulation of the high-frequency dynamics of three-dimensional beam trusses

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The theory of microlocal analysis shows that the energy density associated with the high-frequency vibrations of a three-dimensional Timoshenko beam satisfies a Liouville-type transport equation. In the present application, the material of the beam is assumed to be isotropic. Its parameters are allowed to vary along the beam axis at length scales much larger than the wavelength of the high-frequency waves traveling in it. Moreover, the curvature and torsion of the beam are accounted for. The first part of the paper focuses on the derivation of the transport model for a single three-dimensional beam. In order to extend this model to beam trusses, the reflection/transmission phenomena of the energy fluxes at junctions of beams are described by power flow reflection/transmission operators in a subsequent part. For numerical simulations, a discontinuous Galerkin finite element method is used on account of the discontinuities of the energy density field at the junctions. Thus, a complete mechanical–numerical modeling of the linear transient dynamics of beam trusses is proposed. It is illustrated by numerical examples highlighting some remarkable features of high-frequency vibrations: The onset of a diffusive regime characterized by energy equipartition rules at late times. Energy diffusion is prompted by the multiple reflection/transmission of waves at the junctions, with possible mode (polarization) conversions. This is the regime applicable to the statistical energy analysis of structural acoustics systems. The main purpose of this research is to develop an effective strategy to simulate and predict the transient response of beam trusses impacted by acoustic or mechanical shocks. © 2011 Acoustical Society of America. [DOI: 10.1121/1.3651819]

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I. INTRODUCTION

Spatial structures are often subjected to impulse loads which induce high-frequency wave propagation. Despite some recent research, the characterization of the transient response to such loads remains a difficult problem.^{1,2} These loads can damage the integrity of the structures or equipment attached to them. For example, the solar panels' unfolding is set off by a pyrotechnic shock leading to high-frequency wave propagation in the panels. Another example is the pyrotechnic cut for the space launcher/satellite separation. Thus, it is necessary to improve the prediction of the high-frequency structural responses with efficient theoretical and numerical models.

This paper focuses on the high-frequency dynamics of three-dimensional beam trusses. In the high-frequency range, classical finite element methods are not applicable on account of the small wavelengths and high modal densities. Semi-analytical methods combining exact traveling wave solutions with refined analyses of the reflection and transmission phenomena at the junctions have been proposed in numerous contributions specialized to beam/waveguide networks; see among others the works of Doyle,³ Guo *et al.*,⁴ or Wilcox *et al.*,⁵ and references therein. Qualitative methods have also been developed by engineers—either global approaches like the statistical energy analysis (SEA)⁶ of structural-acoustic systems, or local approaches like the vibrational conductivity

analogy.⁷ The main point of issue concerning the existing local approaches is their extension to complex structures, e.g., beam frames. The original SEA analysis was subsequently improved and made more rigorous in numerous publications, notably by wave approaches. The purpose of this paper is not to review them all. For further information, though, the present work may be related to some extent to several recent contributions based on a wave analysis.^{1,8–13}

The best known and oldest technique of analysis of high-frequency waves is the method of geometrical optics, or ray method.¹⁴ It has been used in structural acoustics and solid mechanics for several decades^{15–17}; see also Ref. 19 for applications to elastic waves in geophysics. It is based on a so-called Wentzel–Kramers–Brillouin–Jeffreys (WKBJ) approximation of the high-frequency, strongly oscillating solutions of the elastic wave equation. In this approximation the wave function is recast as an exponential function with either a real or a complex phase.^{14–19} A typical difficulty of the WKBJ ansatz is raised by the (nonlinear) eikonal equation obtained for the real phase function. This nonlinearity precludes the superposition of different wave phases. The method also yields the formation of caustics even for simple structures.¹⁹ These concerns may be alleviated using a complex phase function, as done for example in a Gaussian beam expansion.^{19–22} Another possibility is to use an alternative partial differential technique based on the Segment Projection Method,²³ which is usable for nonconvex domain. This last method has been applied in the context of geometrical optics for scalar waves. Thus, no polarization and no

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conversion phenomena are considered. A modal approach was also developed by Keller and Rubinov,²⁴ and applied to coupled Timoshenko beams with several types of joints by Chen and Zhou²⁵ and Coleman and Schaffer.²⁶ This method gives asymptotic estimates of the eigenvalues for multi-dimensional scattering problems. The results are valid even in a relatively low frequency range but give no information on the transient response. The latter is considered in Ref. 27, for example, introducing low-order models of the structure dynamics based on a proper orthogonal decomposition. Another approach, which is valid at arbitrarily small wavelengths, is to adopt an energy transport formulation. It is the theory developed here.

Based on microlocal analysis, it can be proved that the energy density associated with high-frequency elastic waves satisfies a Liouville-type transport equation.^{28–31} This theory can be adapted to quantum waves, or classical acoustic and electromagnetic waves as shown in Refs. 28–31. Transport equations can be solved numerically by various methods including Monte Carlo methods³² or nodal/spectral discontinuous “Galerkin” (DG) finite element methods.³³ Both techniques have the advantage to be weakly dissipative and weakly dispersive, depending on the order of interpolation. Thus, they are well suited for late time simulations in order to exhibit the diffusive limit of the transport equations.^{30,34} These methods have been applied to assemblies of two-dimensional beams or plates,³⁵ as well as randomly heterogeneous shells.^{36,37} The main objectives of this paper are twofold. First, the two-dimensional model of high-frequency energy evolution derived by Savin³⁵ is extended to a curved, three-dimensional beam model for application to multi-bay trusses. Second, a general framework for numerical simulations by discontinuous finite elements is derived. This mathematical-numerical model of the transient dynamics of beam trusses shall be able to predict, for example, their possible diffusive behavior at late times or the energy paths within the different bays. It should be apparent that the model is clearly reminiscent of the various wave approaches evoked previously.^{1,3–5,8–22} The main difference is that it describes the propagation of energetic quantities, though the transport equation exhibits basically the same features as the underlying wave equation. In particular, it can be solved exactly for homogeneous infinite media. The analysis of the power flow reflection/transmission processes at the beam junctions is, in addition, very much comparable to the approaches developed in the aforementioned works. The proposed method may then be related to the semi-analytical methods among them, for example, except that propagation phenomena are resolved numerically. It is also concerned with the transient response of the system.

The paper is organized as follows. In Sec. II, the mechanical model of a curved, three-dimensional Timoshenko beam is outlined. The transport theory is introduced in Sec. III and then applied to the Timoshenko beam model. In order to extend the theory to high-frequency wave propagation in multi-bay trusses, the various reflection/transmission phenomena at junctions of beams have to be analyzed. This is the purpose of Sec. IV of the paper. The numerical resolution of the transport equations by a discontinuous finite element

method is considered in a subsequent section. This strategy is illustrated with an application example. The last section offers a few conclusions.

II. THREE-DIMENSIONAL TIMOSHENKO BEAM MODEL

Application of the transport theory first requires one to establish the dynamic behavior of a three-dimensional Timoshenko beam, i.e., its dynamic and constitutive equations. The purpose of the following is to derive these equations prior to the high-frequency setting proposed in Sec. III.

A. Geometry and kinematical hypotheses

The vibrational behavior of the beam is modeled by Timoshenko’s beam kinematics. This model is more relevant in the high-frequency range than the Euler–Bernoulli one. As explained by several authors (see, e.g., Ref. 38, and references therein), Euler–Bernoulli beam kinematics is not valid when the transverse dimensions are not negligible with respect to the wavelength. Indeed, it allows the propagation of energy at speeds approaching infinity in this range.^{39–41} Timoshenko’s model alleviates this shortcoming by including the effects of rotary inertia and shear distortion.⁴¹ It can be seen as a first-order Taylor expansion in the radial coordinate of the motion with respect to the beam neutral fiber (defined in the following). Another important point in this theory is the introduction of a shear coefficient to account for the variation of the shear stress along the cross section. The latter is assumed to be rigid and orthogonal to the neutral fiber at rest. The rigidity assumption is valid in practice for slender, thin beams or rods, and also for moderately thick beams made from some special structural materials.⁴² In this paper, beams’ materials are assumed to be isotropic but not necessarily homogeneous. The material and geometrical parameters vary along the beam axes at a length scale much larger than the wavelength of the excitations imposed on the structure. The implication of this assumption will be apparent in the high-frequency setting of Sec. III A [see Eq. (23)]. In addition, the material parameters are supposed to be symmetrically distributed on the cross section, and the geometrical and inertial centers of the latter are assumed to be superimposed. They form the neutral fiber of the beam, denoted by $\mathcal{C}$. The model to follow allows for curvature and torsion of the neutral fiber. Let $\hat{t}(s)$, $\hat{n}(s)$, and $\hat{b}(s)$ be the tangent, normal, and bi-normal unit vectors constituting a Frenet–Serret frame defined about $\mathcal{C}$. It is parameterized by the curvilinear abscissa s in a (bounded or not) subset $\mathcal{S}$ of $\mathbb{R}$. The corresponding curvilinear coordinates are $s = (s, s_2, s_3)^T$ such that a point x in the subdomain Ω of $\mathbb{R}^3$ occupied by the beam is parameterized by $x = \Psi(s)$ in a fixed reference frame. Here and in the remainder of the paper the superscript $(\cdot)^T$ stands for the transpose of a vector or a matrix. Ω is also written $\Omega = \mathcal{C} \times \Sigma(s)$, where Σ is the beam cross section. Let $x_c(s) = \Psi(s, 0, 0)$ be the position of the neutral fiber, then $\hat{t}(s) = dx_c/ds$, $\hat{n}(s) = |d\hat{t}/ds|^{-1} d\hat{t}/ds$, and $\hat{b}(s) = \hat{t} \times \hat{n}$. The curvature $\kappa(s)$ is defined by $d\hat{t}/ds = \kappa\hat{n}$, and the torsion $\tau(s)$ is defined by $d\hat{b}/ds = -\tau\hat{n}$. The cross section dimensions have to be small in regard to both the

curvature and the torsion.⁴³ Figure 1 shows an example of a beam with curvature but no torsion.

By Timoshenko's beam kinematic assumption, the displacement of a point s of the cross section is written in the local frame:

$$\mathbf{u}(s, t) = \mathbf{u}_c(s, t) + \boldsymbol{\theta}(s, t) \times \mathbf{s}_\perp, \quad (1)$$

where $\mathbf{s}_\perp = \mathbf{s} - (\mathbf{s} \cdot \hat{\mathbf{t}})\hat{\mathbf{t}} \in \Sigma(s)$, $\mathbf{u}_c(s, t) = u_c(s, t)\hat{\mathbf{t}}(s) + v_c(s, t)\hat{\mathbf{n}}(s) + w_c(s, t)\hat{\mathbf{b}}(s)$ is the displacement of the neutral fiber, and $\boldsymbol{\theta}(s, t) = \theta_1(s, t)\hat{\mathbf{t}}(s) + \theta_2(s, t)\hat{\mathbf{n}}(s) + \theta_3(s, t)\hat{\mathbf{b}}(s)$ is the small rotation vector of the cross section. Assumption (1) corresponds to a first-order Taylor expansion of the displacement field with respect to the transverse coordinates $\mathbf{s}_\perp$. It may be enriched to get more accurate for high frequencies by adding higher-order terms in the expansion. The latter would describe the local deformations of the cross sections. For example, additional degrees of freedom associated with the warping deformation of a rod can be introduced,⁴³ adding second-order (and higher) terms in Eq. (1). A third curvature associated with the twist of the beam at rest may also be considered.⁴⁴

B. Equations of motion

Let us consider a curved beam of material volume density $\rho(s)$ and cross-section area $S(s) = \int_\Sigma d\Sigma$. Integrating the local dynamic equilibrium equation in Ω on the cross section Σ and assuming that no external force is applied to the beam, the balance of momentum for the resultant (net) forces is

$$\frac{\partial \mathbf{f}}{\partial s} = \int_\Sigma \rho \ddot{\mathbf{u}} d\Sigma. \quad (2)$$

Here $\mathbf{f}(s, t) = T_1(s, t)\hat{\mathbf{t}}(s) + T_2(s, t)\hat{\mathbf{n}}(s) + T_3(s, t)\hat{\mathbf{b}}(s)$ is the net force acting on the cross section, T_1 is the normal force along $\hat{\mathbf{t}}$, T_2 is the tangent force along $\hat{\mathbf{n}}$, and T_3 is the tangent force along $\hat{\mathbf{b}}$. The double dots $\ddot{\mathbf{u}}$ stand for double derivation with respect to time of $\mathbf{u}$. Assuming that no external couple is applied to the beam either, the balance of angular momentum for the resultant (net) moments is

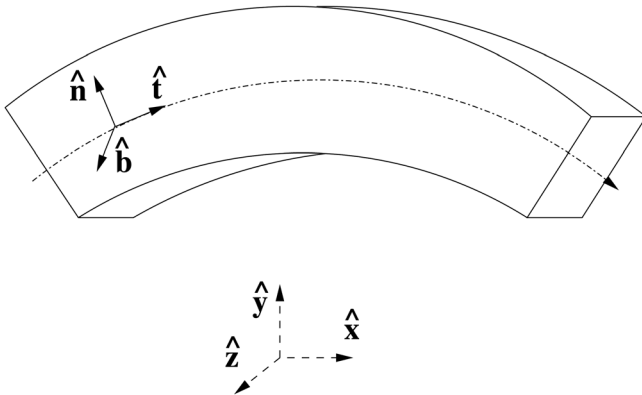


FIG. 1. Curved beam oriented by its local Frenet-Serret frame (full line), with the tangent ($\hat{\mathbf{t}}$), normal ($\hat{\mathbf{n}}$), and bi-normal axes ($\hat{\mathbf{b}}$) and the reference frame (dashed line). The neutral fiber is the dashed-dotted line and the curvature is about the bi-normal axis solely.

$$\frac{\partial \mathbf{m}}{\partial s} + \hat{\mathbf{t}} \times \mathbf{f} = \int_\Sigma (\mathbf{s}_\perp \times \rho \ddot{\mathbf{u}}) d\Sigma. \quad (3)$$

Here, $\mathbf{m}(s, t) = M_1(s, t)\hat{\mathbf{t}}(s) + M_2(s, t)\hat{\mathbf{n}}(s) + M_3(s, t)\hat{\mathbf{b}}(s)$ is the net moment acting on the cross section, M_1 is the torsional moment, M_2 is the bending moment along $\hat{\mathbf{n}}$, and M_3 is the bending moment along $\hat{\mathbf{b}}$. Now the derivation of the Frenet-Serret frame $[\hat{\mathbf{t}}(s), \hat{\mathbf{n}}(s), \hat{\mathbf{b}}(s)]$ with respect to the curvilinear abscissa s along the neutral fiber is given by the so-called Frenet-Serret formulas. Plugging Eq. (1) into Eq. (2) yields the first set of equations of motion:

$$\mathbf{D}\mathbf{F} = \rho_s \ddot{\mathbf{U}}_c, \quad (4)$$

where $\rho_s(s) = \int_\Sigma \rho d\Sigma$, and

$$\mathbf{D}(s) = \begin{bmatrix} \frac{\partial}{\partial s} & -\kappa & 0 \\ \kappa & \frac{\partial}{\partial s} & -\tau \\ 0 & \tau & \frac{\partial}{\partial s} \end{bmatrix} \quad (5)$$

is the so-called Frenet-Serret matrix operator. Plugging Eq. (1) into Eq. (3) yields the second set of equations of motion:

$$\mathbf{D}\mathbf{M} + \hat{\mathbf{t}} \times \mathbf{F} = \mathbf{J}[\rho] \ddot{\boldsymbol{\Theta}}, \quad (6)$$

where $\mathbf{J}[\rho](s) = \text{diag}(I[\rho](s), J[\rho](s), K[\rho](s))$, $J[\rho] = \int_\Sigma s_2^2 \rho d\Sigma$, $K[\rho] = \int_\Sigma s_3^2 \rho d\Sigma$, and $I[\rho] = J[\rho] + K[\rho]$ are the inertias of the beam. In Eqs. (4) and (6), capital letters $\mathbf{U}_c$, $\boldsymbol{\Theta}$, $\mathbf{F}$, and $\mathbf{M}$ stand equivalently for the vectors in $\mathbb{R}^3$ constituted by the coordinates of $\mathbf{u}_c$, $\boldsymbol{\theta}$, $\mathbf{f}$, and $\mathbf{m}$, respectively, in the local frame $(\hat{\mathbf{t}}, \hat{\mathbf{n}}, \hat{\mathbf{b}})$; for example, $\mathbf{F}(s, t) = (T_1(s, t), T_2(s, t), T_3(s, t))^T$ and $\mathbf{M}(s, t) = (M_1(s, t), M_2(s, t), M_3(s, t))^T$.

C. Resultant constitutive equations

The derivation of the constitutive equations is adapted from the analysis proposed by Huang.⁴³ It is based on the computation of the internal work, which is independent of the computational frame. The internal work $V(s, t)$ is

$$V = - \int_S [\mathbf{D}\mathbf{F} \cdot \mathbf{U}_c + (\mathbf{D}\mathbf{M} + \hat{\mathbf{t}} \times \mathbf{F}) \cdot \boldsymbol{\Theta}] ds. \quad (7)$$

Integrating by parts on $]0, L[\subset \mathcal{S}$ yields

$$V = \int_0^L [\mathbf{F} \cdot (\mathbf{D}\mathbf{U}_c + \hat{\mathbf{t}} \times \boldsymbol{\Theta}) + \mathbf{M} \cdot \mathbf{D}\boldsymbol{\Theta}] ds - [\mathbf{F} \cdot \mathbf{U}_c + \mathbf{M} \cdot \boldsymbol{\Theta}]_{s=0}^{s=L}. \quad (8)$$

The boundary conditions prescribe either T_1 or u_c , either T_2 or v_c , either T_3 or w_c , either M_1 or θ_1 , either M_2 or θ_2 , and either M_3 or θ_3 . By definition, the net force acting on the cross section Σ is

$$\mathbf{f} = \int_{\Sigma} \boldsymbol{\sigma} \hat{\mathbf{t}} d\Sigma, \quad (9)$$

where $\boldsymbol{\sigma}(s, t)$ is the second-order stress tensor within the medium Ω . Also the net moment acting on Σ is

$$\mathbf{m} = \int_{\Sigma} \mathbf{s}_{\perp} \times \boldsymbol{\sigma} \hat{\mathbf{t}} d\Sigma. \quad (10)$$

Substituting Eqs. (9) and (10) into Eq. (8) leads to

$$V = \int_0^L \int_{\Sigma} [\boldsymbol{\sigma} \hat{\mathbf{t}} \cdot (\mathbf{D}\mathbf{U}_c + \hat{\mathbf{t}} \times \boldsymbol{\Theta}) + (\mathbf{s}_{\perp} \times \boldsymbol{\sigma} \hat{\mathbf{t}}) \cdot \mathbf{D}\boldsymbol{\Theta}] d\Sigma ds \quad (11)$$

for the internal work in $]0, L[$. The latter may be written as

$$V = \int_0^L \int_{\Sigma} \boldsymbol{\sigma} : \boldsymbol{\epsilon} d\Sigma ds, \quad (12)$$

where $\boldsymbol{\epsilon}(s, t)$ is the second-order strain tensor, $\mathbf{G}:\mathbf{H} := \text{Tr}(\mathbf{G}\mathbf{H}^T)$ is the tensor scalar product of the second-order tensors $\mathbf{G}$ and $\mathbf{H}$, and Tr is the trace of a square matrix. Factorizing the terms with respect to σ_{1j} in Eq. (11) allows one to identify ϵ_{1j} from Eq. (12). In an elastic medium and under the assumption of small displacements, the stress-strain relation is Hooke's law given by

$$\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\epsilon}. \quad (13)$$

$\mathbf{C}(s)$ is the relaxation tensor depending on the material's Young's modulus $E(s)$ and shear modulus $G(s)$ if it is isotropic. This constitutive law is privileged in the following, although the analysis could be extended to orthotropic materials for example. These results will be presented in future communications. Substituting the Hooke's law [Eq. (13)] with the expression of the strain ϵ_{1j} given by Eqs. (11) and (12) into Eq. (9) finally leads to

$$\mathbf{F} = SC_1(\mathbf{D}\mathbf{U}_c + \hat{\mathbf{t}} \times \boldsymbol{\Theta}). \quad (14)$$

where $SC_1(s) := \text{diag}\{\int_{\Sigma} E d\Sigma, \mu(\int_{\Sigma} G d\Sigma) \mathbf{I}_2\}$ is the relaxation tensor integrated over the cross section, with μ a shear reduction factor and $\mathbf{I}_n$ is the $n \times n$ identity matrix. Substituting Eq. (13) into Eq. (10) also leads to

$$\mathbf{M} = \mathbf{C}_2 \mathbf{D}\boldsymbol{\Theta}, \quad (15)$$

where $\mathbf{C}_2(s) := \text{diag}(\mu J[G](s), J[E](s), K[E](s))$ is the relaxation tensor for rotational motions integrated over Σ . Other authors, e.g., Tsay and Kingsbury⁴⁵ or Treysède,⁴⁶ have found similar results using alternative methods.

D. Energetic observables

The dynamics of the beam reduced to its neutral fiber $s \in \mathcal{S}$ by integration over the cross section $\Sigma(s)$ is described by Eqs. (4), (6), (14), and (15). Introducing the state vector

$\mathbf{X} = (\dot{\mathbf{U}}_c^T, \dot{\boldsymbol{\Theta}}^T, \mathbf{F}^T, \mathbf{M}^T)^T$ of $\mathbb{R}^{12}$, they can be written as a 12×12 first-order hyperbolic system:

$$\begin{cases} \mathbf{A}(s) \partial_t \mathbf{X} = \mathbf{P}(s, \partial_s) \mathbf{X} \\ \mathbf{X}(0, s) = \mathbf{X}_0(s) \end{cases}, \quad (16)$$

where $\mathbf{A}$ is the block diagonal matrix $\mathbf{A}(s) = \text{diag}(\rho_s \mathbf{I}_3, \mathbf{J}[\rho], (SC_1)^{-1}, \mathbf{C}_2^{-1})$. Here, $\mathbf{P}(s, \partial_s)$ is the operator:

$$\mathbf{P}(s, \partial_s) = \begin{bmatrix} 0 & 0 & \mathbf{D} & 0 \\ 0 & 0 & \boldsymbol{\Omega} & \mathbf{D} \\ \mathbf{D} & \boldsymbol{\Omega} & 0 & 0 \\ 0 & \mathbf{D} & 0 & 0 \end{bmatrix}, \quad \boldsymbol{\Omega} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}. \quad (17)$$

It may also be written $\mathbf{P}(s, \partial_s) = \mathbf{D}^1 \partial_s + \mathbf{D}^0(s)$, where $\mathbf{D}^1$ and $\mathbf{D}^0$ are 12×12 matrices derived straightforwardly from Eqs. (17) and (5). Note that $\mathbf{D}^1$ is symmetric and has elements equal to 0 or 1, and $\mathbf{D}^0$ is skew-symmetric. Here $\mathbf{X}_0(s)$ is a vector of initial conditions. Then the mechanical energy density $\mathcal{E} \in \mathbb{R}_+$ for the beam motion is

$$\mathcal{E}(s, t) = \frac{1}{2} (\mathbf{X}, \mathbf{X})_A, \quad (18)$$

and the power flow density $\Pi \in \mathbb{R}$ within the beam is

$$\Pi(s, t) = -\frac{1}{2} (\mathbf{X}, \mathbf{X})_{D^1} = -(\mathbf{F} \cdot \dot{\mathbf{U}}_c + \mathbf{M} \cdot \dot{\boldsymbol{\Theta}}), \quad (19)$$

where the notation $(\mathbf{u}, \mathbf{v})_{\mathbf{G}} := \mathbf{u}^T \mathbf{G} \mathbf{v}$ is introduced for a square matrix $\mathbf{G}$. Finally, taking the scalar product of Eq. (16) with $\mathbf{X}$ leads to the energy conservation law, or continuity equation:

$$\partial_t \mathcal{E} + \partial_s \Pi = 0. \quad (20)$$

III. HIGH-FREQUENCY WAVE PROPAGATION IN A THREE-DIMENSIONAL BEAM

This section focuses on the computation of the solution of system (16) in the high-frequency regime. High-frequency vibrations of the beam may be generated by initial conditions oscillating at a small length scale ε proportional to the wavelength. The high-frequency regime thus corresponds to $\varepsilon \rightarrow 0$. These initial conditions may be chosen for example as plane waves in the form $\mathbf{X}_\varepsilon^0(s) = \boldsymbol{\chi}(s) e^{iks/\varepsilon}$, where $k \in \mathbb{R}$ and $\boldsymbol{\chi}(s)$ is a slowly varying vector. As easily seen, it is not possible to describe the limit of $\mathbf{X}_\varepsilon^0$ as $\varepsilon \rightarrow 0$. By the hyperbolicity of Eq. (16) its solution $\mathbf{X}_\varepsilon(s, t)$ has the same shape as the initial condition, it is also not possible to describe its high-frequency limit. However, the energy input by the initial condition remains finite and will propagate in the beam. One should be able to describe and track it by Eq. (20), provided its limit as $\varepsilon \rightarrow 0$ is known. Therefore, energetic quantities associated with $\mathbf{X}_\varepsilon$, or observables proportional to $|\mathbf{X}_\varepsilon|^2$ in some sense, shall be considered in this very limit. The link between the solution $\mathbf{X}_\varepsilon$ of Eq. (16) and these observables as $\varepsilon \rightarrow 0$ is established using a Wigner transform of $\mathbf{X}_\varepsilon$ and its limit as $\varepsilon \rightarrow 0$, the so-called Wigner

measure. The theory developed by Papanicolaou and Ryzhik³⁰ or Guo and Wang³¹ outlines how the limit is obtained, and how a continuity equation of form (20) is derived for the Wigner measure. The main novelty in the proposed analysis is that all energetic quantities shall be resolved in phase space (position $\times$ wave vector) in order to correctly build up their high-frequency limits and evolution properties. The general framework is described in Sec. III A, and its application to a curved three-dimensional Timoshenko beam is given in Sec. III B.

A. High-frequency energy density and transport properties

In this section, the high-frequency limit of a first-order hyperbolic system of form (16) is considered in a general setting. First, the Wigner transform and its properties are introduced. This transform is used to estimate the energy density associated with the solutions of the system of Eq. (16). Let $X(s, t)$ be a complex function defined on $\mathbb{R}_s \times \mathbb{R}_t$ with values in $\mathbb{C}^n$. Provided that X is square integrable with respect to s , its spatial Wigner transform is defined by

$$W_\varepsilon[X](s, k, t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ik \cdot s'} X\left(s - \varepsilon \frac{s'}{2}, t\right) \times X^*\left(s + \varepsilon \frac{s'}{2}, t\right) ds', \quad (21)$$

where $X^* = \overline{X}^T$ stands for the conjugate transpose of X . The $n \times n$ matrix $W_\varepsilon[X]$ is Hermitian and not necessarily positive, but it does become positive in the high-frequency limit $\varepsilon \rightarrow 0$. Further, this (weak) limit always exists for a square integrable function. The Wigner transform also has the following property:

$$\int_{\mathbb{R}} W_\varepsilon[X](s, k, t) dk = X(s, t) X^*(s, t), \quad (22)$$

hence its link with the energy if one takes the trace of the above-presented equation. Now let us consider a vector X_ε in $\mathbb{C}^n$ satisfying the first-order hyperbolic problem:

$$\begin{cases} \varepsilon A(s) \partial_t X_\varepsilon = P_\varepsilon(s, \varepsilon \partial_s) X_\varepsilon, \\ X_\varepsilon(0, s) = X_\varepsilon^0(s). \end{cases} \quad (23)$$

where $P_\varepsilon(s, \varepsilon \partial_s) = D^1 \varepsilon \partial_s + \varepsilon D^0(s)$. This system is obtained from Eq. (16) by multiplying it with ε , assuming in addition that $A(s)$ and $D^0(s)$ are independent of ε . This assumption was introduced earlier in Sec. II A. It means that the parameters of the materials constituting the beam and its curvatures vary on a scale much longer than the small wavelength ε imposed by the initial condition X_ε^0 . Indeed, the initial data X_ε^0 vary on a scale lower or equal to ε , which is the case of the plane waves $X_\varepsilon^0(s) \propto e^{iks/\varepsilon}$ considered earlier. More generally, these data have to satisfy the so-called ε -oscillatory strong condition stating that $(|\varepsilon \partial_s X_\varepsilon^0|)^2$ remains at least locally integrable on $\mathbb{R}$.²⁹ High-frequency plane waves are only a particular case satisfying this condition. Finally,

A and D^1 are also assumed to be symmetric so that system (23) is said to be symmetric. Then provided that the sequence (X_ε) remains bounded in the set of square integrable functions with respect to $s \in \mathbb{R}$, its Wigner transform $W_\varepsilon[X_\varepsilon]$ has an Hermitian weak limit (denoted by W) as $\varepsilon \rightarrow 0$, which is also non-negative. This very limit is the so-called Wigner measure of the sequence (X_ε) .^{28,29} The energy density of Eq. (18) is expressed in terms of W by

$$\mathcal{E}(s, t) = \frac{1}{2} \int_{\mathbb{R}} A : W dk, \quad (24)$$

and the power flow density (19) is

$$\Pi(s, t) = -\frac{1}{2} \int_{\mathbb{R}} D^1 : W dk \quad (25)$$

as $\varepsilon \rightarrow 0$. As the sequence (X_ε) satisfies system (23), it can be shown in addition that its Wigner measure can be written as^{30,31,36}

$$W(s, k, t) = \sum_{\alpha=1}^M \sum_{i,j=1}^{R_\alpha} w_\alpha^{ij}(s, k, t) v_{\alpha_i}(s, k) \otimes v_{\alpha_j}(s, k). \quad (26)$$

In the above-presented equation, $(v_{\alpha_i})_{1 \leq i \leq R_\alpha}$ are the eigenvectors of the dispersion matrix $\Gamma(s, k) := kA^{-1}(s)D^1$ associated with the eigenvalue λ_α of which order of multiplicity is R_α . These orders of multiplicity are assumed to be independent of $(s, k) \in \mathcal{S} \times \mathbb{R}$ and are such that $\sum_{\alpha=1}^M R_\alpha = n$. The $R_\alpha \times R_\alpha$ coherence matrices W_α , defined as $[W_\alpha]_{ij} = w_\alpha^{ij}$, satisfy the transport equations

$$\partial_t W_\alpha + \{\lambda_\alpha, W_\alpha\} + [W_\alpha, N_\alpha] = 0, \quad (27)$$

where $\{g, h\} = \partial_k g \partial_s h - \partial_s g \partial_k h$ is the usual Poisson's bracket, $[G, H] = GH + H^*G^*$, and N_α is an $R_\alpha \times R_\alpha$ coupling matrix with elements:

$$[N_\alpha(s)]_{ij} = (v_{\alpha_j}, P(s, \partial_s) v_{\alpha_i}) - \partial_s \lambda_\alpha (\partial_k v_{\alpha_i}, v_{\alpha_j})_A - \frac{1}{2} (\partial_s \partial_k \lambda_\alpha) \delta_{ij}. \quad (28)$$

B. Application to a three-dimensional Timoshenko beam

Let $k := k\hat{t}$ be the wave vector within a beam oriented by its tangent unit vector $\hat{t}$. As D^0 is rescaled as a first-order term in ε , it does not influence the eigenvalues of the dispersion matrix Γ of system (16) of Sec. II D. Thus, the curvatures have no influence on the high-frequency vibrations of the beam. The eigenvalues of Γ are

$$\begin{aligned} \lambda_P^\pm(s, k) &= \pm c_P(s) |k| = \pm \sqrt{\frac{E}{\rho}} |k| \quad \text{each of multiplicity 3} \\ \lambda_T^\pm(s, k) &= \pm c_T(s) |k| = \pm \sqrt{\frac{\mu G}{\rho}} |k| \quad \text{each of multiplicity 3,} \end{aligned} \quad (29)$$

where c_P and c_T are the group velocities of longitudinal and transverse waves, respectively (as these waves are

non-dispersive in the high-frequency range, the group velocities are also the phase velocities). Hence $R_\alpha^\pm = 3$ for $\alpha = P, T$ and either the forward (+) or the backward (−) waves. The associated eigenvectors, orthonormal with respect to the inner product $(\mathbf{u}, \mathbf{v})_T$, are

$$\begin{aligned} \mathbf{v}_{P_1}^\pm &= \left(\frac{\text{sign}(k)}{\sqrt{2\rho_S}}, 0, 0, 0, 0, \pm \sqrt{\frac{ES}{2}}, 0, 0, 0, 0 \right)^T \\ \mathbf{v}_{P_2}^\pm &= \left(0, 0, 0, 0, \frac{\text{sign}(k)}{\sqrt{2J[\rho]}}, 0, 0, 0, 0, \pm \sqrt{\frac{J[E]}{2}}, 0 \right)^T \\ \mathbf{v}_{P_3}^\pm &= \left(0, 0, 0, 0, 0, \frac{\text{sign}(k)}{\sqrt{2K[\rho]}}, 0, 0, 0, 0, \pm \sqrt{\frac{K[E]}{2}} \right)^T \\ \mathbf{v}_{T_1}^\pm &= \left(0, \frac{\text{sign}(k)}{\sqrt{2\rho_S}}, 0, 0, 0, 0, \pm \sqrt{\frac{\mu GS}{2}}, 0, 0, 0, 0 \right)^T \\ \mathbf{v}_{T_2}^\pm &= \left(0, 0, \frac{\text{sign}(k)}{\sqrt{2\rho_S}}, 0, 0, 0, 0, \pm \sqrt{\frac{\mu GS}{2}}, 0, 0, 0 \right)^T \\ \mathbf{v}_{T_3}^\pm &= \left(0, 0, 0, \frac{\text{sign}(k)}{\sqrt{2I[\rho]}}, 0, 0, 0, 0, \pm \sqrt{\frac{\mu I[G]}{2}}, 0, 0 \right)^T. \end{aligned} \quad (30)$$

The eigenvectors associated with the eigenvalues $P^\pm$ are the compressional (1) and bending [(2) and (3)] energy modes, and the eigenvectors associated with the eigenvalues $T^\pm$ are the shear [(1) and (2)] and torsional (3) energy modes. As these eigenvectors are linearly independent, no coupling occurs between modes in the high-frequency range away from the boundaries and interfaces. Their influence on the coupling is analyzed in the next section. The 3×3 coherence matrices $\mathbf{W}_\alpha^\pm$, $\alpha = P, T$, satisfy the Liouville-type transport equations:

$$\partial_t \mathbf{W}_\alpha^\pm \pm \text{sign}(k) c_\alpha \partial_s \mathbf{W}_\alpha^\pm \mp |k| \partial_s c_\alpha \partial_k \mathbf{W}_\alpha^\pm = \mathbf{0}. \quad (31)$$

Indeed it can be checked that the brackets $[\mathbf{W}_\alpha^\pm, \mathbf{N}_\alpha^\pm]$ vanish for beams (in fact, the coupling matrices $\mathbf{N}_\alpha^\pm$ do so). These results extend to three dimensions the results of Savin³⁵ for two-dimensional beam trusses. Accordingly, it can be observed that $\mathbf{W}_\alpha^-(s, k, t) = \mathbf{W}_\alpha^+(s, -k, t)$ for $\alpha = P, T$, so that the space time energy density reduces to

$$\mathcal{E}(s, t) = \sum_{\alpha=P,T} \int_{\mathbb{R}} \text{Tr} \mathbf{W}_\alpha^+(s, k, t) dk. \quad (32)$$

Similarly, the power flow density vector $\mathbf{\Pi}(s, t) := \mathbf{\Pi}(s, t) \hat{\mathbf{t}}(s)$ in the local frame is

$$\begin{aligned} \mathbf{\Pi}(s, t) &= \sum_{\alpha=P,T} c_\alpha(s) \int_{\mathbb{R}} \text{Tr} \mathbf{W}_\alpha^+(s, k, t) \hat{\mathbf{k}} dk \\ &= \sum_{\alpha=P,T} \sum_{j=1}^{R_\alpha} \int_{\mathbb{R}} \pi_\alpha^{ij}(s, k, t) dk. \end{aligned} \quad (33)$$

Here $\pi_\alpha^{ij}(s, k, t) := c_\alpha(s) w_\alpha^{ij}(s, k, t) \hat{\mathbf{k}}$ is the power flow density of a mode $\alpha \in \{P, T\}$, $1 \leq j \leq R_\alpha$, in a beam oriented by its tangent unit vector $\hat{\mathbf{t}}$, with $\hat{\mathbf{k}} := \mathbf{k}/|\mathbf{k}| = \text{sign}(k) \hat{\mathbf{t}}$. If $k > 0$ the

energy flux travels in the same direction as $\hat{\mathbf{t}}$, but if $k < 0$ it travels in the opposite direction. For convenience the sign function is denoted by $\text{sign}(k) = \hat{k}$ in the remainder of the paper. The energy and power flow densities [Eqs. (32) and (33)] to be computed ultimately depend solely on the diagonal elements w_α^{ij} of $\mathbf{W}_\alpha$. Transport equations (31) do not couple them, although the boundary/interface conditions considered in the next section do so. Thus, only Eq. (31) for the elements $(w_\alpha^{ij})^+$ of $\mathbf{W}_\alpha^+$ has to be considered, reminding the symmetry $\mathbf{W}_\alpha^+(s, k, t) = \mathbf{W}_\alpha^-(s, -k, t)$. Introducing the set of energy modes $E = \{P_1, P_2, P_3, T_1, T_2, T_3\}$ corresponding to the eigenvectors $\mathbf{v}_{P_j}^\pm$ and $\mathbf{v}_{T_j}^\pm$, $1 \leq j \leq 3$, and the eigenvalues λ_P^+ and λ_T^+ (counted with their orders of multiplicity), the set of transport equations to be considered is finally reduced to

$$\partial_t w_\alpha + c_\alpha \hat{k} \partial_s w_\alpha - |k| \partial_s c_\alpha \partial_k w_\alpha = 0, \quad \alpha \in E. \quad (34)$$

Accordingly, the power flow density of a mode $\alpha \in E$ is denoted by

$$\pi_\alpha(s, \mathbf{k}, t) = c_\alpha(s) w_\alpha(s, k, t) \hat{\mathbf{k}}. \quad (35)$$

IV. HIGH-FREQUENCY POWER FLOW REFLECTION/TRANSMISSION COEFFICIENTS IN BEAM ASSEMBLIES

The high-frequency energy density propagation in a beam is described by transport equations (34). In order to use this model for complex three-dimensional beam trusses, the reflection/transmission processes at the junctions have to be described. They contribute to couple the energy densities w_α . This section deals with the energy density propagation in a junction of two beams, and then an arbitrary number $\mathcal{N}$ of beams. Power flow reflection/transmission coefficients are derived along the same lines as in Savin³⁵ for two-dimensional junctions. The subsequent analysis considers the basic case where incident waves are traveling in a single beam and reflected/transmitted in the other beams connected to the junction. Incidences in several or all members of the junction may be treated by superposition of this basic situation.

A. Energy transport in coupled structures

1. A junction of two beams

Let us consider two beams occupying the domains Ω_1 and Ω_2 of $\mathbb{R}^3$. They are parameterized by their curvilinear coordinates $\mathbf{s} = (s, s_2, s_3)^T \in \mathbb{R}^3$ such that $s \in \mathcal{S}_1 \subset \mathbb{R}$ for beam 1 and $s \in \mathcal{S}_2 \subset \mathbb{R}$ for beam 2. Here $\mathcal{S}_p$, $p = 1, 2$, is an interval of $\mathbb{R}$ such that $\mathcal{C}_p = \Psi_p(\mathcal{S}_p)$ is the curved line in $\mathbb{R}^3$ constituting the neutral fiber of beam p . Their junction $\Gamma := \partial\Omega_1 \cap \partial\Omega_2$ reduces to a single point $\mathbf{x}_0 = \Psi_p(s_0^p)$ in this parameterization. Let $\hat{\mathbf{t}}^p$ be the tangent vector to $\mathcal{C}_p$, $p = 1, 2$, pointing outward from beam p at the junction. Then it is assumed for convenience that s_0^p may be written $s_0^p = s_0 \hat{\mathbf{t}}^p$ in the local Frenet frame of each beam p . The Liouville equations (34) hold within each beam. However, their Hamiltonians $\lambda_\alpha^p(s, \mathbf{k}) := c_\alpha^p(s) |\mathbf{k}|$ shall be preserved across the interface as in phase space description of geometrical optics^{47,48} where they remain constant along the (energy)

rays. This applies even if they are reflected or transmitted by that interface. Here c_α^p stands for the different group velocities of longitudinal ($\alpha = P_j$, $1 \leq j \leq 3$) and transverse ($\alpha = T_j$, $1 \leq j \leq 3$) waves in beam p . Thus, one has

$$\lambda_\alpha^p(s_0, \mathbf{k}) = \lambda_\beta^q(s_0, \mathbf{k}) := \omega, \quad \forall p, q = 1, 2, \quad \forall \alpha, \beta \in E. \quad (36)$$

Condition (36) holds for either transmitted ($p \neq q$) or reflected ($p = q$) waves. It can be used to determine the reflected and transmitted wavenumbers $|k_\alpha^p|$ from either side of the interface given an incident wavenumber $|k|$. Moreover, power flow (33) is conserved across the junction owing to the Rankine–Hugoniot condition written for transport equations (34).⁴⁹ For example, the energy density traveling in beam 1 and impinging the junction at s_0 with an incident wave vector $\mathbf{k}$ such that $\mathbf{k} \cdot \hat{\mathbf{t}}^1 > 0$ is partially reflected and partially transmitted, whereas the overall power flow remains constant. This property reads

$$\pi_\alpha^1(s_0, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^1 = \sum_{\beta \in E} \left(\left| \pi_\beta^1(s_0, \mathbf{k}_\beta^1, t) \cdot \hat{\mathbf{t}}^1 \right| + \left| \pi_\beta^2(s_0, \mathbf{k}_\beta^2, t) \cdot \hat{\mathbf{t}}^2 \right| \right), \quad (37)$$

where the left traces of π_α in beam p at the junction are denoted by

$$\pi_\alpha^p(s_0, \mathbf{k}, t) := \lim_{h \downarrow 0} \pi_\alpha(s_0 \cdot \hat{\mathbf{t}}^p - h, \mathbf{k}, t). \quad (38)$$

Here $c_\alpha^1|\mathbf{k}| = c_\beta^p|\mathbf{k}_\beta^p| \forall \alpha, \beta \in E$ by Eq. (36), and $\mathbf{k}_\beta^1 = -|\mathbf{k}_\beta^1|\hat{\mathbf{t}}^1$, $\mathbf{k}_\beta^2 = -|\mathbf{k}_\beta^2|\hat{\mathbf{t}}^2$. The first and second terms on the right-hand side of Eq. (37) correspond to the fluxes flowing away from the junction in beams 1 and 2, respectively, after reflection and transmission. They read as follows:

$$\left| \pi_\alpha^p(s_0, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^p \right| = \sum_{\beta \in E} \left[\rho_{\alpha\beta}^{pp}(s_0) \pi_\beta^p(s_0, \mathbf{k}_\beta^p, t) \cdot \hat{\mathbf{t}}^p + \tau_{\alpha\beta}^{pq}(s_0) \pi_\beta^q(s_0, \mathbf{k}_\beta^q, t) \cdot \hat{\mathbf{t}}^q \right], \quad (39)$$

for $p \neq q \in \{1, 2\}$ and $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^p < 0$, with $\mathbf{k}_\beta^1 = |\mathbf{k}_\beta^1|\hat{\mathbf{t}}^1$ and $\mathbf{k}_\beta^2 = |\mathbf{k}_\beta^2|\hat{\mathbf{t}}^2$. $\rho_{\alpha\beta}^{pp}(s_0)$ and $\tau_{\alpha\beta}^{pq}(s_0)$ are power flow reflection/transmission coefficients of the junction, respectively. They are also called efficiencies in the dedicated literature. The above-presented relations constitute the boundary/interface conditions to be used to solve the transport equations (34). It should be noted that they remain valid everywhere within the beams if the convention $\rho_{\alpha\beta}^{pp} = 0$ and $\tau_{\alpha\beta}^{pq} = \delta_{\alpha\beta}$ is adopted for all $s \neq s_0$.

2. A junction of $\mathcal{N}$ beams

Now this analysis can be generalized straightforwardly to a junction of $\mathcal{N} \geq 2$ beams. Equation (39) for the fluxes

flowing away from that junction in beam p after reflection and transmission reads

$$\left| \pi_\alpha^p(s_0, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^p \right| = \sum_{\beta \in E} \left[\rho_{\alpha\beta}^{pp}(s_0) \pi_\beta^p(s_0, \mathbf{k}_\beta^p, t) \cdot \hat{\mathbf{t}}^p + \sum_{\substack{q=1 \\ q \neq p}}^{\mathcal{N}} \tau_{\alpha\beta}^{pq}(s_0) \pi_\beta^q(s_0, \mathbf{k}_\beta^q, t) \cdot \hat{\mathbf{t}}^q \right] \quad (40)$$

for $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^p < 0$, where $\mathbf{k}_\beta^q = |\mathbf{k}_\beta^q|\hat{\mathbf{t}}^q$ such that $c_\alpha^p(s_0)|\mathbf{k}_\alpha^p| = c_\beta^q(s_0)|\mathbf{k}_\beta^q| := \omega$ for $1 \leq p, q \leq \mathcal{N}$, $\alpha, \beta \in E$ by Eq. (36). One may also define linear reflection operators $\mathcal{R}_{\alpha\beta}^{pp} : w_\beta \mapsto w_\alpha$ and transmission operators $T_{\alpha\beta}^{pq} : w_\beta \mapsto w_\alpha$ provided that Eq. (40) is written in terms of the energy densities w_α rather than the flow densities π_α :

$$\left| \pi_\alpha^p(s_0, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^p \right| = \sum_{\beta \in E} \left[\mathcal{R}_{\alpha\beta}^{pp}(s_0) c_\beta^p w_\beta^p(s_0, \mathbf{k}_\beta^p, \hat{\mathbf{t}}^p, t) + \sum_{\substack{q=1 \\ q \neq p}}^{\mathcal{N}} T_{\alpha\beta}^{pq}(s_0) c_\beta^q w_\beta^q(s_0, \mathbf{k}_\beta^q, \hat{\mathbf{t}}^p, t) \right] \quad (41)$$

for $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^p < 0$. Here the same definition as for the flow density, Eq. (38), is used for the traces of c_α and w_α .

B. Computation of reflection/transmission coefficients for a beam junction

This section outlines how the power flow reflection/transmission coefficients $\rho_{\alpha\beta}^{pp}(s_0)$ and $\tau_{\alpha\beta}^{pq}(s_0)$ may be computed. Some examples are presented in order to illustrate these results.

1. Theoretical analysis

A junction of $\mathcal{N}$ beams is considered. It is geometrically described by the Euler angles between beam p and beam q . They are denoted by ψ_{pq} , θ_{pq} , and φ_{pq} with the usual convention $(\hat{\mathbf{n}}, \hat{\mathbf{b}}, \hat{\mathbf{t}})$ meaning that the composition of three intrinsic rotations about the moving frame axes is used to rotate the local Frenet frame of beam p to the local frame of beam q . This very rotation is denoted by $\mathbf{R}^{qp}$. The curvatures and torsions are ignored as they have no influence on the high-frequency regime, as shown in Sec. III B). The junction is at $s = s_0$, and it is assumed without loss of generality in the remaining that $s_0 = 0$. It is also considered that waves are incident in beam 1. Figure 2 shows an example of a junction of three beams, and the existing waves in each beam. From the analysis in Sec. III B it is concluded that the high-frequency wavenumber for compressional and bending motions is $k_p^p = \omega/c_p^p$, where ω is the circular frequency of Eq. (36). Similarly, the high-frequency wavenumber for torsional and shear motions is $k_T^p = \omega/c_T^p$ (see, e.g., Ref. 35). The incident waves, traveling in beam 1 in the direction of increasing s may be written as a linear superposition of

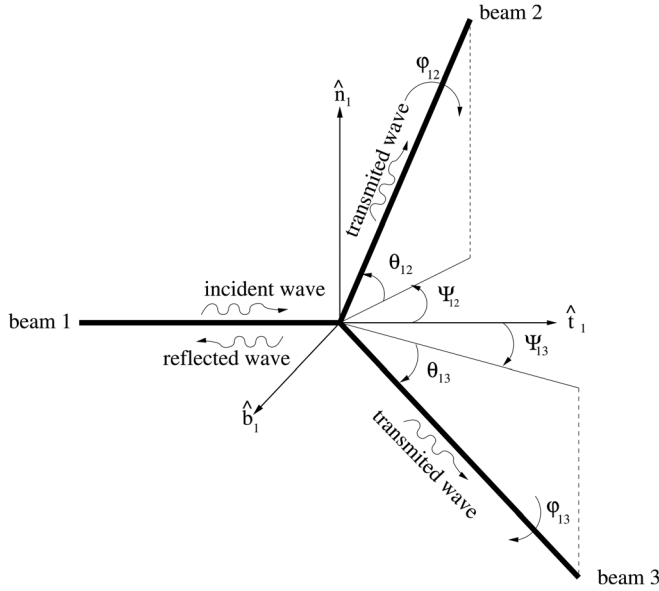


FIG. 2. A junction of three coupled beams with traveling waves in each beam. The junction is defined by three Euler angles: The first one is ψ_{1i} , the second one is θ_{1i} , and the last one is ϕ_{1i} using the usual convention $(\hat{n}, \hat{b}, \hat{t})$ for the junction between beam 1 and beam i , $i = 2, 3$.

$$\begin{aligned}
 u_c^I &= e^{-ik_p^1 s} && \text{for a pure longitudinal motion along } \hat{t}, \\
 v_c^I &= e^{-ik_t^1 s} && \text{for a pure shear motion along } \hat{n}, \\
 w_c^I &= e^{-ik_t^1 s} && \text{for a pure shear motion along } \hat{b}, \\
 \theta_1^I &= -ik_t^1 e^{-ik_t^1 s} && \text{for a pure torsional motion about } \hat{t}, \\
 \theta_2^I &= -ik_p^1 e^{-ik_p^1 s} && \text{for a pure bending motion about } \hat{n}, \\
 \theta_3^I &= -ik_p^1 e^{-ik_p^1 s} && \text{for a pure bending motion about } \hat{b}.
 \end{aligned}$$

The corresponding reflected waves traveling in beam 1 in direction of decreasing s read as follows:

$$\begin{aligned}
 u_c^R &= B_1 e^{ik_p^1 s}, \\
 v_c^R &= B_2 e^{ik_t^1 s}, \\
 w_c^R &= B_3 e^{ik_t^1 s}, \\
 \theta_1^R &= ik_t^1 B_4 e^{ik_t^1 s}, \\
 \theta_2^R &= ik_p^1 B_5 e^{ik_p^1 s}, \\
 \theta_3^R &= ik_p^1 B_6 e^{ik_p^1 s}.
 \end{aligned}$$

Accordingly, the transmitted waves traveling in beam q in direction of increasing s for that beam read as

$$\begin{aligned}
 u_c^{Tq} &= C_1^q e^{-ik_p^q s}, \\
 v_c^{Tq} &= C_2^q e^{-ik_t^q s},
 \end{aligned}$$

$$w_c^{Tq} = C_3^q e^{-ik_t^q s},$$

$$\theta_1^{Tq} = -ik_t^q C_4^q e^{-ik_t^q s},$$

$$\theta_2^{Tq} = -ik_p^q C_5^q e^{-ik_p^q s},$$

$$\theta_3^{Tq} = -ik_p^q C_6^q e^{-ik_p^q s}.$$

Coefficients B_i and C_i^q are obtained from the continuity of displacements, rotations, forces, and moments at the junction:

$$\begin{aligned}
 U_c^I(0) + U_c^R(0) &= \mathbf{R}^{1q} U_c^{Tq}(0) \\
 \Theta^I(0) + \Theta^R(0) &= \mathbf{R}^{1q} \Theta^{Tq}(0) \\
 \mathbf{F}^I(0) &= \sum_{q=2}^N \mathbf{R}^{1q} \mathbf{F}^q(0) \\
 \mathbf{M}^I(0) &= \sum_{q=2}^N \mathbf{R}^{1q} \mathbf{M}^q(0)
 \end{aligned} \tag{42}$$

$\mathbf{F}^q$ and $\mathbf{M}^q$ correspond to the net force and net moment in beam q , respectively. $\mathbf{R}^{1q}$ is the rotation matrix corresponding to the projection of the frame of beam q on the frame of beam 1. Inserting constitutive equations (14) and (15) into Eq. (42) yields a linear system in terms of displacements and rotations solely. It can be easily solved for each type of incident wave and allows one to compute the coefficients B_i , C_i^q . To be consistent with transport model (31) governing the phase space energy densities, the associated power flows are derived from the generic formula for the energy flux of a high-frequency plane wave propagating in a beam.³⁵ The transient power flow in beam q being

$$\Pi^q = -\Re\{\mathbf{F}^q \cdot \overline{\mathbf{U}}_c^q + \mathbf{M}^q \cdot \overline{\Theta}^q\},$$

where $\Re\{\mathcal{Z}\}$ denotes the real part of $\mathcal{Z}$, the time average power flow associated to a plane wave is

$$\langle \Pi^q \rangle = \frac{1}{2} \Re\left\{ i\omega (\mathbf{F}^q \cdot \overline{\mathbf{U}}_c^q + \mathbf{M}^q \cdot \overline{\Theta}^q) \right\}$$

The high-frequency power reflection/transmission coefficients for coupled Timoshenko beams are then defined by

$$\rho_{\alpha\beta}^{11} = \frac{\langle \Pi_\alpha^R \rangle}{\langle \Pi_\beta^I \rangle}, \tag{43}$$

$$\tau_{\alpha\beta}^{q1} = \frac{\langle \Pi_\alpha^{Tq} \rangle}{\langle \Pi_\beta^I \rangle}, \tag{44}$$

respectively. Here $\langle \Pi_\alpha^R \rangle$ is the time average power flow for the reflected waves corresponding to the mode $\alpha \in E$, $\langle \Pi_\alpha^{Tq} \rangle$ is the time average power flow for the transmitted wave corresponding to the mode α in beam q , and $\langle \Pi_\beta^I \rangle$ is the time average power flow for an incident wave in the mode β .

2. Examples

It may be observed by computation that, in the high-frequency limit, the reflection/transmission coefficients corresponding to translation incident motions u_c^l , v_c^l , or w_c^l and rotational reflected/transmitted motions θ_i^R , θ_i^{Tq} , $i = 1, 2$, or 3, are null. The coefficients related to rotational incident motions and translational reflected/transmitted motions vanish as well. All other coefficients are independent of the circular frequency ω . Figures 3 and 4 display the reflection/transmission coefficients computed for two Timoshenko beams ($\mathcal{N} = 2$) with $E_2 = 2E_1$, all other parameters being identical ($\nu_1 = \nu_2$, $\rho_1 = \rho_2$, and $\mathbf{J}_1 = \mathbf{J}_2$). The Euler angles between the beams at the junction are $-\pi/2 \leq \psi_{12} \leq \pi/2$, $\theta_{12} = \pi/4$, and $\varphi_{12} = 0$. Obviously, the sum of the power flow reflection/transmission coefficients is equal to one for a given angle. This property expresses energy conservation. Figures 5 and 6 display the reflection/transmission coefficients for three Timoshenko beams ($\mathcal{N} = 3$) with the same geometrical and mechanical parameters ($E_1 = E_2 = E_3$, $\nu_1 = \nu_2 = \nu_3$, $\rho_1 = \rho_2 = \rho_3$, and $\mathbf{J}_1 = \mathbf{J}_2 = \mathbf{J}_3$). The Euler angles between the beams at the junction are $-\pi/2 \leq \psi_{12} \leq \pi/2$, $\theta_{12} = \pi/4$, $\varphi_{12} = 0$, $\psi_{13} = 0$, $\theta_{13} = 0$, and $\varphi_{13} = 0$. Again, the sum of the power flow reflection/transmission coefficients is equal to one for a given angle.

V. NUMERICAL SIMULATION

Discontinuities in the velocity fields at the junction do not allow one to use classical finite element method to solve numerically the Liouville equations (34) with the interface condition of Eq. (37) or Eq. (40). Indeed, although the overall power flow density is continuous across the junction owing to the Rankine–Hugoniot condition, the energy density itself does not necessarily have to be continuous across that junction. Therefore a DG finite element method^{33,49,50} is used to solve Eq. (34) together with the reflection/transmission conditions of Eq. (40). The theoretical background of the DG method is recalled in Sec. V A, whereas some numerical implementation issues are addressed in Sec. V B. An illustrative example is presented in Sec. V C.

A. DG finite elements

The DG method for transport equations (34) is first written on a single beam $s \in \mathcal{S}$. Let us consider the subdivision $\mathcal{T}_h = \cup_{r=1}^K D_r$ of $\mathcal{S}$ (which is now assumed to be bounded) into K non-overlapping subdomains, or finite elements D_r . These elements are defined by $D_r = [s_{r-1/2}, s_{r+1/2}]$. The largest element size of this mesh is $h = \max_{1 \leq r \leq K} |s_{r+1/2} - s_{r-1/2}|$. It is assumed that the velocities c_α are constant over the elements; they are denoted by c_α^r . Here kl is a fixed parameter,

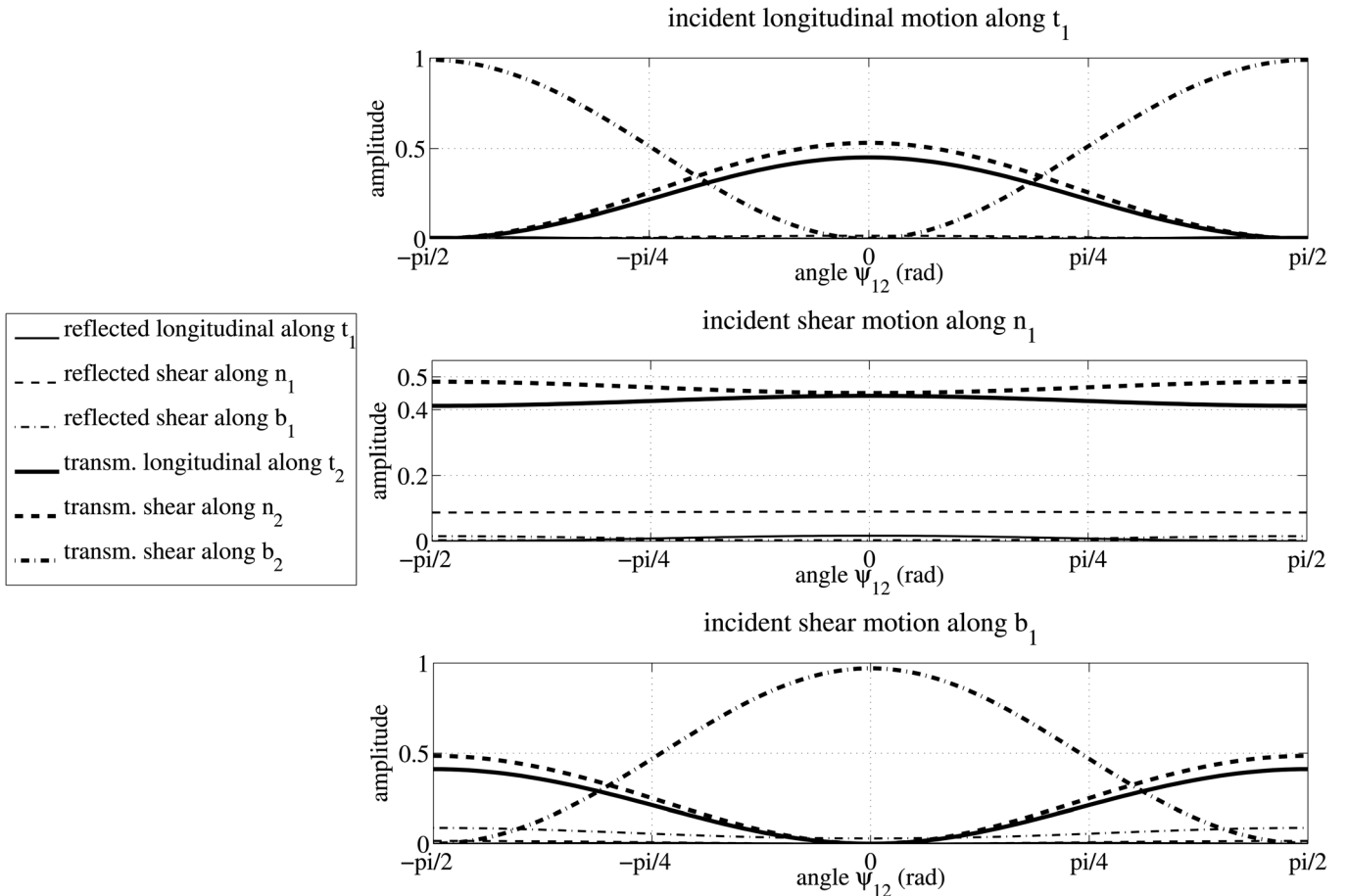


FIG. 3. Evolution of the power flow reflection/transmission coefficients as functions of the angle ψ_{12} for a junction of two Timoshenko beams with $E_2 = 2E_1$, $\nu_1 = \nu_2$, $\rho_1 = \rho_2$, and $\mathbf{J}_1 = \mathbf{J}_2$ for translational incident waves (compressional and shear motions). The Euler angles at the junction are $-\pi/2 \leq \psi_{12} \leq \pi/2$, $\theta_{12} = \pi/4$, and $\varphi_{12} = 0$.

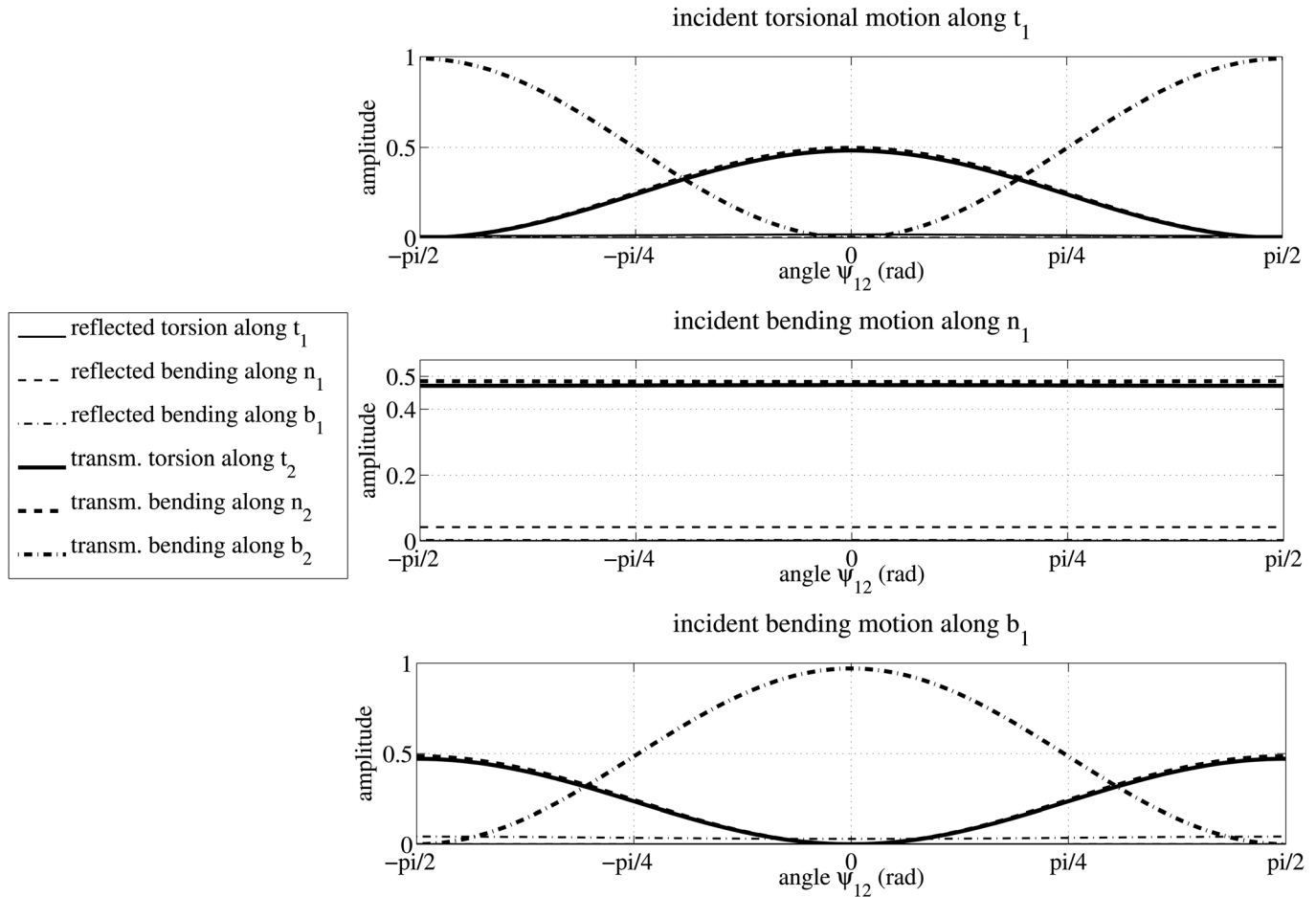


FIG. 4. Evolution of the power flow reflection/transmission coefficients as functions of the angle ψ_{12} for a junction of two Timoshenko beams with $E_2 = 2E_1$, $\nu_1 = \nu_2$, $\rho_1 = \rho_2$, and $\mathbf{J}_1 = \mathbf{J}_2$ for rotational incident waves (torsional and bending motions). The Euler angles at the junction are $-\pi/2 \leq \psi_{12} \leq \pi/2$, $\theta_{12} = \pi/4$, and $\phi_{12} = 0$.

and the w_α 's are considered as functions of $\hat{k}$ rather than k . A Galerkin variational formulation of Eq. (34) in an *ad hoc* functional set $\mathbf{V}_h$ defined on $\mathcal{T}_h$ is as follows: Find $w_\alpha \in \mathbf{V}_h$, $\alpha \in E$ such that

$$\int_{D_r} \left[\frac{\partial w_\alpha}{\partial t}(s, \hat{k}, t) v(s) - c_\alpha^r \hat{k} w_\alpha(s, \hat{k}, t) \frac{dv}{ds} \right] ds = -[c_\alpha^r \hat{k} w_\alpha v]_{s_{r-1/2}}^{s_{r+1/2}} = -[\pi_\alpha \cdot \hat{\mathbf{t}}^r v]_{s_{r-1/2}}^{s_{r+1/2}}, \quad \forall v \in \mathbf{V}_h. \quad (45)$$

The power flow densities $\pi_\alpha(s, \hat{k}, t)$ in the above-presented formulation are defined by Eq. (35), and $\hat{\mathbf{t}}^r$ is the outward unit normal to $\partial D_r = \{s_{r-1/2}, s_{r+1/2}\}$. As the sought solutions w_α of Eq. (45) may be discontinuous at a junction of beams, or at the interfaces between elements, the global space $\mathbf{V}_h$ is defined as $\mathbf{V}_h = \bigoplus_{r=1}^K \mathbf{V}_h^r$, where the locally defined test spaces $\mathbf{V}_h^r \subseteq H^1(D_r)$ do not enforce any particular boundary conditions on ∂D_r . As a consequence of the lack of such conditions for the local solution and the test functions, the former is *a priori* multiply defined at the interfaces between elements (a junction being only one of them). Thus the boundary flux $\pi_\alpha \cdot \hat{\mathbf{t}}^r$ in Eq. (45) is not defined. That is why it is replaced by a numerical flux π_α^* , which depends on the traces of w_α from both sides of an

interface between elements. This leads to the following scheme:

$$\int_{D_r} \left[\frac{\partial w_\alpha}{\partial t}(s, \hat{k}, t) v(s) - c_\alpha^r \hat{k} w_\alpha(s, \hat{k}, t) \frac{dv}{ds} \right] ds = -[\pi_\alpha^* \cdot \hat{\mathbf{t}}^r v]_{s_{r-1/2}}^{s_{r+1/2}}, \quad \forall v \in \mathbf{V}_h. \quad (46)$$

The numerical flux π_α^* at the interface $s = s_{r+1/2}$, say, is constructed from the computed values of $w_\alpha(s_{r+1/2}, \hat{k}, t)$ in $\mathbf{V}_h^r$ (the trace of w_α at $s_{r+1/2}$ in D_r) and $w_\alpha(s_{r+1-1/2}, \hat{k}, t)$ in $\mathbf{V}_h^{r+1}$ (the trace of w_α at $s_{r+1/2}$ in D_{r+1}) so as to be consistent and conservative. Let

$$\pi_\alpha^*(s_{r+1/2}) := \pi^*(s_{r+1/2}; (c_\alpha w_\alpha)^r, (c_\alpha w_\alpha)^{r+1}) \quad (47)$$

for $v^r|_{s_{r+1/2}} \equiv v(s_{r+1/2})$ in $\mathbf{V}_h^r$ and $v^{r+1}|_{s_{r+1/2}} \equiv v(s_{r+1-1/2})$ in $\mathbf{V}_h^{r+1}$, recalling definition (38). Then consistency reads

$$\pi^*(s_{r+1/2}; c_\alpha v, c_\alpha v) = c_\alpha v \hat{\mathbf{k}}|_{s=s_{r+1/2}},$$

meaning that true energy flux (35) is recovered when π^* is considered with a continuous function. As for conservation it reads as follows:

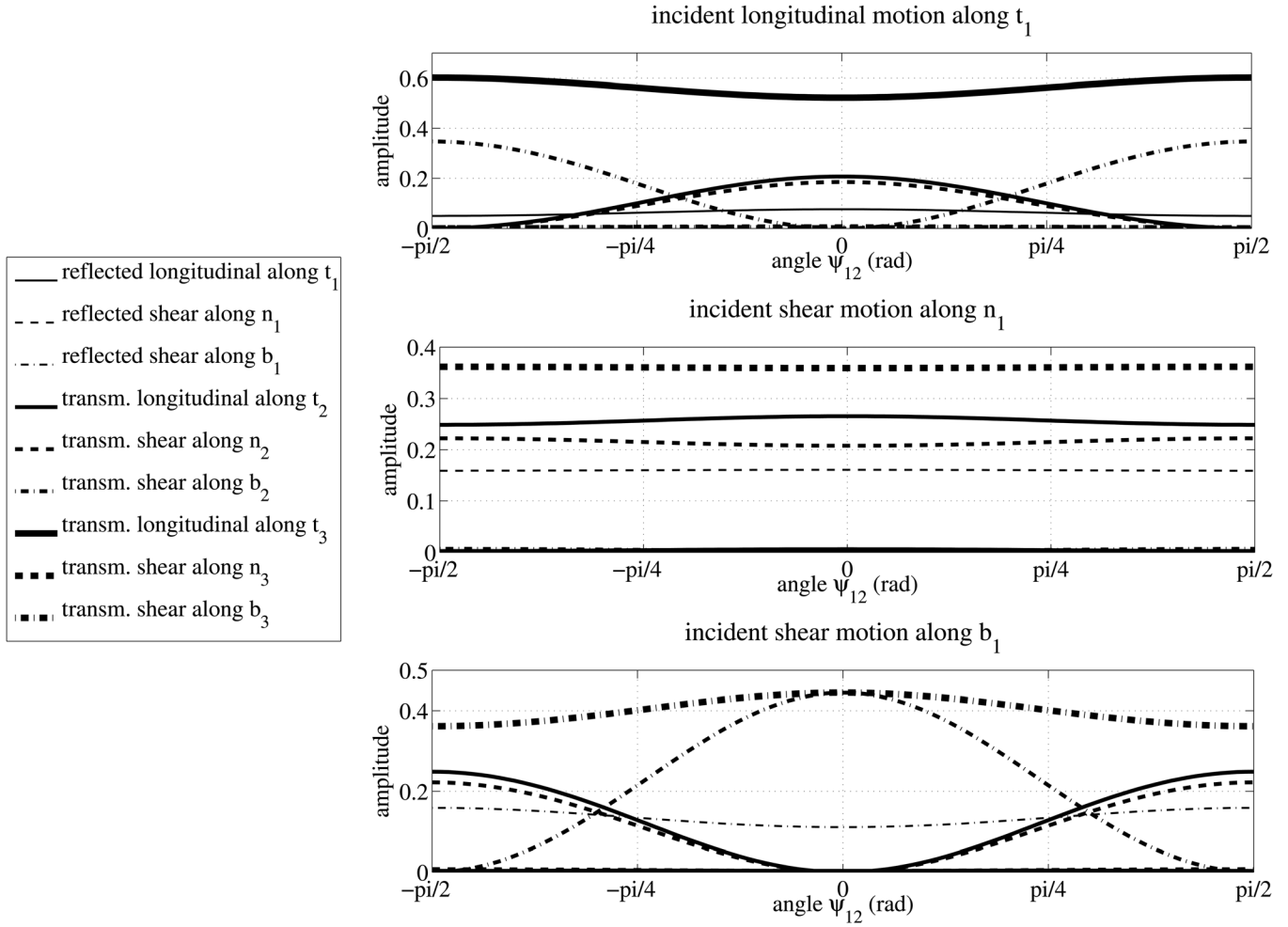


FIG. 5. Evolution of the power flow reflection/transmission coefficients as functions of the angle ψ_{12} for a junction of three identical Timoshenko beams for translational incident waves (compressional and shear motions). The Euler angles between the beams at the junction are $-\pi/2 \leq \psi_{12} \leq \pi/2$, $\theta_{12} = \pi/4$, $\varphi_{12} = 0$, $\psi_{13} = \theta_{13} = 0$, and $\varphi_{13} = 0$.

$$\pi^*(s_{r+1/2}; v^r, v^{r+1}) \cdot \hat{\mathbf{t}}^r + \pi^*(s_{r+1-1/2}; v^r, v^{r+1}) \cdot \hat{\mathbf{t}}^{r+1} = 0$$

at $\{s_{r+1/2}\} = \partial D_r \cap \partial D_{r+1}$.

B. Numerical implementation

Three main ingredients are needed to complete the numerical implementation of the DG method once the subdivision $\mathcal{T}_h$ has been performed: The definition of the numerical flux function π^* on all elements, the choice of the approximation set V_h , and the choice of a time-integration scheme for Eq. (46).

1. Numerical fluxes

The numerical fluxes ensuring the stability and consistency of the discontinuous Galerkin method for transport equations have the form:^{49,50}

$$\pi^*(s_{r+1/2}; v^r, v^{r+1}) = \frac{1}{2}(v^r, v^{r+1})\hat{\mathbf{k}} + A_r(v^r - v^{r+1})\hat{\mathbf{t}}^r,$$

where A_r is a positive scalar. For example, the upwind flux corresponds to $A_r = (1/2)|\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^r|$ and the Lax–Friedrichs flux

corresponds to $A_r = 1/2$. Invoking Eq. (41), this expression is generalized as follows for an interface (a junction) $\{s_{r+1/2}\} = \{s_{r'-1/2}\} = \partial D_r \cap \partial D_{r'}$, $r \neq r' \in \mathcal{I}_r$, where the group velocities c_α are discontinuous: For $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^r < 0$,

$$\pi_\alpha^*(s_{r+1/2}, t) \cdot \hat{\mathbf{t}}^r = \sum_{\beta \in E} \left[\mathcal{R}_{\alpha\beta}^{rr'}(c_\beta w_\beta)^r(s_{r+1/2}, \mathbf{k}_\beta^r \cdot \hat{\mathbf{t}}^r, t) + \sum_{r' \in \mathcal{I}_r} T_{\alpha\beta}^{rr'}(c_\beta w_\beta)^{r'}(s_{r'-1/2}, \mathbf{k}_\beta^{r'} \cdot \hat{\mathbf{t}}^{r'}, t) \right] \quad (48)$$

and for $\hat{\mathbf{k}} \cdot \hat{\mathbf{t}}^r > 0$,

$$\pi_\alpha^*(s_{r+1/2}, t) \cdot \hat{\mathbf{t}}^r = \pi_\alpha^r(s, \mathbf{k}, t) \cdot \hat{\mathbf{t}}^r. \quad (49)$$

$\mathcal{I}_r$ is the set of indices of the beam elements that are connected to the beam element r . This choice corresponds to an upwind numerical flux whenever $\mathcal{R}_{\alpha\beta}^{rr'} \equiv 0$ and $T_{\alpha\beta}^{rr'} \equiv \delta_{\alpha\beta}$ at any point s which is not a beam junction, in which case $\mathcal{I}_r = \{r+1\}$.

2. Approximation set

The approximation set V_h is constructed as follows:

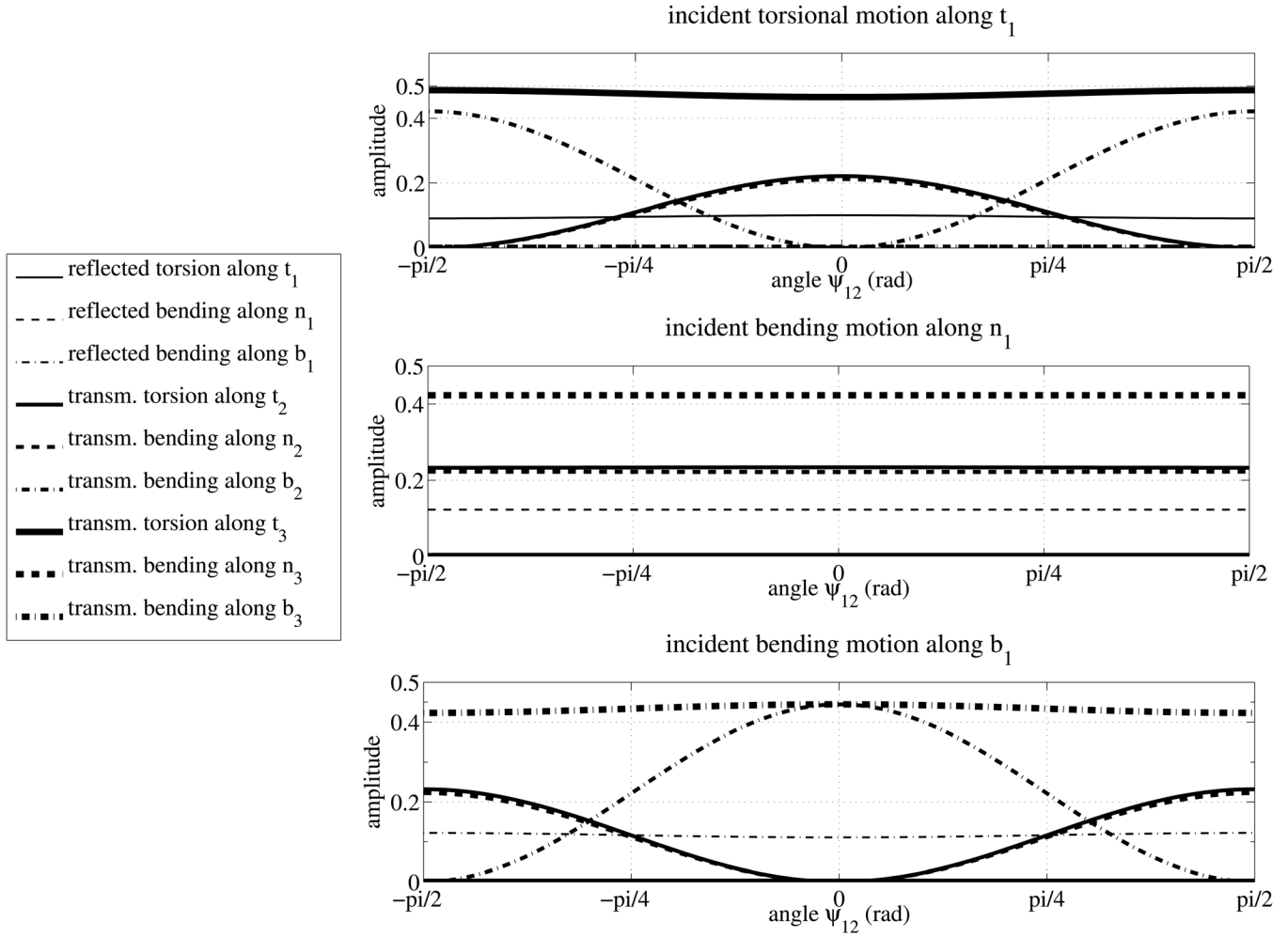


FIG. 6. Evolution of the power flow reflection/transmission coefficients as functions of the angle ψ_{12} for a junction of three identical Timoshenko beams for rotational incident waves (torsional and bending motions). The Euler angles between the beams at the junction are $-\pi/2 \leq \psi_{12} \leq \pi/2$, $\theta_{12} = \pi/4$, $\varphi_{12} = 0$, $\psi_{13} = \theta_{13} = 0$, and $\varphi_{13} = 0$.

$$V_h = \left\{ v|_{D_r} \in \mathbb{P}_\ell(D_r), 1 \leq r \leq K \right\}, \quad (50)$$

where $\mathbb{P}_\ell(D_r) \subset H^1(D_r)$ is typically a polynomial set. In our numerical simulations $\mathbb{P}_\ell(D_r)$ is spanned by a set of orthonormal Legendre polynomials L_j , $0 \leq j \leq \ell$, up to the order ℓ . The approximate solution w_α^h of Eq. (46) reads as follows:

$$w_\alpha^h(s, \hat{k}, t)|_{D_r} = \sum_{j=0}^{\ell} W_{\alpha j}(t, \hat{k}) L_j^r(s), \quad \hat{k} = \pm 1.$$

where $L_j^r(s) = L_j[2(s - s_r)/\Delta s_r]$, $\Delta s_r = s_{r+1/2} - s_{r-1/2}$, and $s_r = (1/2)(s_{r-1/2} + s_{r+1/2})$. Let $\mathbf{W}^r(t) = \{W_{\alpha j}^r(t, \hat{k})\}_{0 \leq j \leq \ell, 1 \leq \alpha \leq M, \hat{k} = \pm 1}$ be the vector in $\mathbb{R}^{2M(\ell+1)}$ of the degrees of freedom for the element D_r . Then Eq. (46) yields

$$M^r \frac{d\mathbf{W}^r}{dt} + K^r \mathbf{W}^r + F^r \mathbf{W}^r + \sum_{r' \in \mathcal{I}_r} F^{r'} \mathbf{W}^{r'} = 0, \quad (51)$$

with an energetic initial condition $\mathbf{W}^r(0)$. The latter is constituted by the coefficients of the projection of the initial

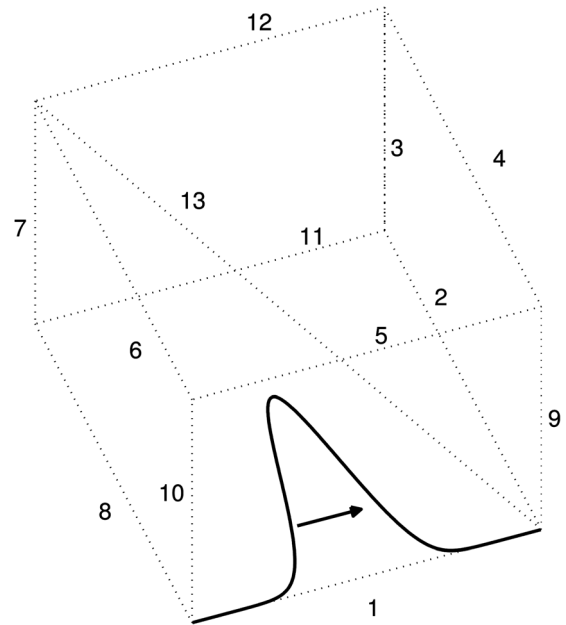


FIG. 7. A view of the geometry of the beam truss (dashed line) and the energetic initial condition (full line) in beam 1. All the beams have the same cross section and the same mechanical parameters. The initial propagation direction (arrow) is $\hat{k} = 1$.

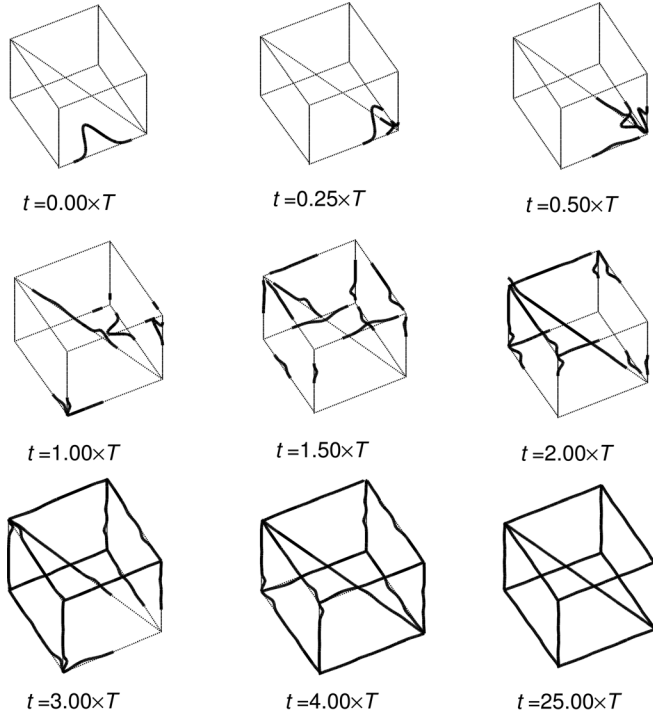


FIG. 8. Evolution of the total energy density $\mathcal{E}(s, t)$ (thick full line) in the beam truss of Fig. 7 (dashed line) for an initial compressional wave. T is defined as the time needed by a transverse wave to travel across beam 1. Only the nonvanishing energy density is displayed. It can be seen that the energy is spread rather uniformly over the entire structure at late times.

conditions for the w_α 's on the Legendre polynomial basis. The local “mass matrix” $\mathbb{M}^r$ is diagonal with

$$\mathbb{M}_{\epsilon\epsilon'}^r = \frac{\Delta s_r}{2} \delta(\epsilon - \epsilon') \mathbf{I}_M \otimes \mathbf{I}_{\ell+1}, \quad \epsilon, \epsilon' = \pm 1,$$

for the forward ($\epsilon = 1$) or backward ($\epsilon = -1$) propagation directions. The local “stiffness” matrix $\mathbb{K}^r$ is

$$\mathbb{K}_{\epsilon\epsilon'}^r = \epsilon \delta(\epsilon - \epsilon') \mathbf{C}^r \otimes \mathbf{K}, \quad \epsilon, \epsilon' = \pm 1$$

where $\mathbf{C}^r = \text{diag}(c_\alpha^r)_{1 \leq \alpha \leq M}$, and the $(\ell + 1) \times (\ell + 1)$ matrix $\mathbf{K}$ has the following entries:

$$K_{jj'} = \int_{-1}^{+1} \frac{dL_j(s)}{ds} L_{j'}(s) ds, \quad 0 \leq j, j' \leq \ell.$$

Here the Kronecker product $\mathbb{Z} = \mathbf{G} \otimes \mathbf{H}$ of an $m \times m$ matrix $\mathbf{G}$ and an $n \times n$ matrix $\mathbf{H}$ is the $mn \times mn$ matrix $\mathbb{Z}$ such that $Z_{(i-1)n+p, (j-1)n+q} = G_{ij} H_{pq}$ for $1 \leq i, j \leq m$ and $1 \leq p, q \leq n$. At last for a junction $\{s_{r+1/2}\} = \{s_{r'-1/2}\} = \partial D_r \cap \partial D_{r'}$ the local flux matrix $\mathbb{F}^r$ is

$$\mathbb{F}_{\epsilon\epsilon'}^r = [\delta(\epsilon - \epsilon') \mathbf{C}^r - \delta(\epsilon + 1) \mathbf{R}^r(s_{r+1/2})] \otimes \mathbf{L}(+1, +1), \quad \epsilon, \epsilon' = \pm 1,$$

and the coupling flux matrices $\mathbb{F}^{r'}$ (coupling the element D_r with its neighbors) are

$$\mathbb{F}_{\epsilon\epsilon'}^{r'} = -\delta(\epsilon + 1) \delta(\epsilon' + 1) \mathbf{T}^{r'}(s_{r'-1/2}) \otimes \mathbf{L}(+1, -1)$$

where $\mathbf{L}(\epsilon, \epsilon') = \mathbf{L}(\epsilon) \otimes \mathbf{L}(\epsilon')$ with $\mathbf{L}(\epsilon) = [L_0(\epsilon), L_1(\epsilon), \dots, L_\ell(\epsilon)]^T$, and

$$R_{\alpha\beta}^r(s_{r+1/2}) = c_\beta^r \mathcal{R}_{\alpha\beta}^{rr}(s_{r+1/2}),$$

$$T_{\alpha\beta}^r(s_{r'-1/2}) = c_\beta^{r'} \mathcal{T}_{\alpha\beta}^{r'r}(s_{r'-1/2})$$

Let $K = h|k|$ and $[z]$ be the integer part of $z \geq 0$. The numerical dispersion and dissipation errors introduced by this spatial discretization are $O(K^{2\ell+3})$ and $O(K^{2\ell+2})$ respectively, for decentered numerical fluxes ($A_r \neq 0$), or $O(K^{4[\ell/2]+3})$ and 0 for centered fluxes ($A_r = 0$).⁵¹ These so-called “superconvergence” properties are very much desirable for long time simulations in order to exhibit the diffusion limit of transport equations (34) with boundary conditions (39).

3. Time integration

Finally, time integration of the semi-discretized system (51) is performed by a strong stability-preserving (SSP) high-order Runge–Kutta scheme.⁵² This fully discretized scheme is referred to as the Runge–Kutta discontinuous Galerkin (RKDG) method in the dedicated literature; see Ref. 33, and references therein. The Runge–Kutta scheme is not reproduced here as it is a direct application of the results expounded in the aforementioned references.

C. Numerical example

A beam truss constituted by $\mathcal{N} = 13$ three-dimensional beams is considered. Its shape is a cube having a side length $L = 5$ unit length, such that $L_p = L$ for $1 < p \leq 12$. The 13th beam is a transverse one linking two opposite vertices of this cube through its diagonal. Thus, its length is $L_{13} = \sqrt{3}L$; see Fig. 7. All beams have the same rectangular cross sections of area S and are made from the same homogeneous, isotropic elastic material. Thus, all beams have the same velocities c_P and c_T for the compressional-bending and shear-torsional modes, respectively. The Poisson ratio is $\nu = 0.3$ and the shear reduction factor is $\mu = 5(1 + \nu)/(6 + 5\nu)$. Transport equations (34) hold in each beam and the reflection/transmission coefficients are directly computed from the approach outlined in Sec. IV B. The forward and backward flows are written as the superposition of reflected and transmitted flows including all existing energy modes, $\alpha = P_j, T_j, j = 1, 2, 3$, as described in Sec. IV A. The energetic initial condition has a Gaussian shape and loads the mode P_1 (load case 1, a pure compressional wave) or the mode T_3 (load case 2, a pure torsional wave). It is applied to, say, beam labeled 1 with a fixed initial direction $\hat{k} = +1$; see again Fig. 7.

The nodal density is fixed at $10/L$ uniform spatial elements per unit length for all beams. Legendre polynomials up to the eighth order ($\ell = 8$) are used as local basis functions. The integration in time is performed by a seventh-order Runge–Kutta SSP scheme. The time scale $T = L/c_T$ is introduced: It is the time needed by a transverse wave to

travel across the side of the cube. This parameter has been set to $T = 25$ s in our simulations. The Courant number $CFL = c_T \Delta t / h$ has been fixed to $CFL = 0.01$. Figure 8 shows the evolution of the overall energy density [Eq. (32)] computed in the beam truss for the load case 1. Only the non-vanishing energy density is displayed on this plot. It can be seen that the energy is spread over the entire structure at late times. Discontinuities of the energy density are also noticeable at the junctions. At last, the observed numerical wave velocities agree with those of the theory. Thus the numerical scheme is practically non-dispersive (the theoretical dispersion error is about 5×10^{-20} for our choice $\ell = 8$, which is much below machine precision). Figures 9 and 11 show the evolution of the total energy $\mathcal{E}^p(t) := \int_0^{L_p} \mathcal{E}(s, t) ds$ for each beam, $1 \leq p \leq 13$. It is first noticed that the energetic behavior of parallel beams is similar at late times. Hence, for the sake of clarity, the results of Figs. 9 and 11 are displayed for each beam group, and then the sums of the energies for each group. The first group is constituted by the beams parallel to the initially loaded beam 1 (beams 1, 5, 10, 11). The second group is constituted by the vertical beams 2, 4, 6, 8. The third group is constituted by the remaining beams 3, 7, 9, 12, whereas the diagonal beam 13 is considered apart. It should be observed that the total energy for the entire truss is virtually conserved. This is a required property of the numerical scheme retained (the theoretical numerical dissipation error being about 5×10^{-19}). Moreover, the total energy in each beam tends to a limit as $t \rightarrow \infty$, which apparently depends

on its mechanical parameters. This phenomenon characterizes the diffusive regime holding at late times.

Finally, Figs. 10 and 12 show the ratio between the overall transverse and longitudinal energy per beam $\mathcal{E}_T^p(t)/\mathcal{E}_P^p(t)$ where

$$\mathcal{E}_\alpha^p(t) = \sum_{j=1}^3 \sum_{\hat{k}=\pm 1} \int_0^{L_p} w_{\alpha_j}(s, \hat{k}, t) ds, 1 \leq p \leq 13, \alpha = P, T.$$

This ratio converges to a constant value of the form $n_T^p c_P / n_P^p c_T$ where n_P^p and n_T^p are the number of longitudinal and transverse modes generated in each beam by the reflection/transmission processes at the junctions. n_P and n_T depend on the initial condition and the shape of the truss. However, this limit should be analyzed theoretically more in detail; this issue is beyond the scope of this paper but it is the subject of ongoing research. For example, the initial motion in Figs. 9 and 10 is a compressional mode $\alpha = P_1$ in beam 1. It is transmitted as transverse shear modes ($\alpha = T_1$ or T_2) in the beams perpendicular to it (beams 2 and 12), and as transverse shear modes and the same compressional mode in beam 13. However, no conversion to bending ($\alpha = P_2$ or P_3) or torsional ($\alpha = T_3$) modes occurs at any time. Thus, $n_P = 1$ and $n_T = 2$ in this case. For Figs. 11 and 12 the initial mode is a pure torsional wave $\alpha = T_3$ hit on beam 1 as well. According to the reflection/transmission coefficients for such a rotational incident wave, $n_P = 2$ and $n_T = 1$ (the transverse

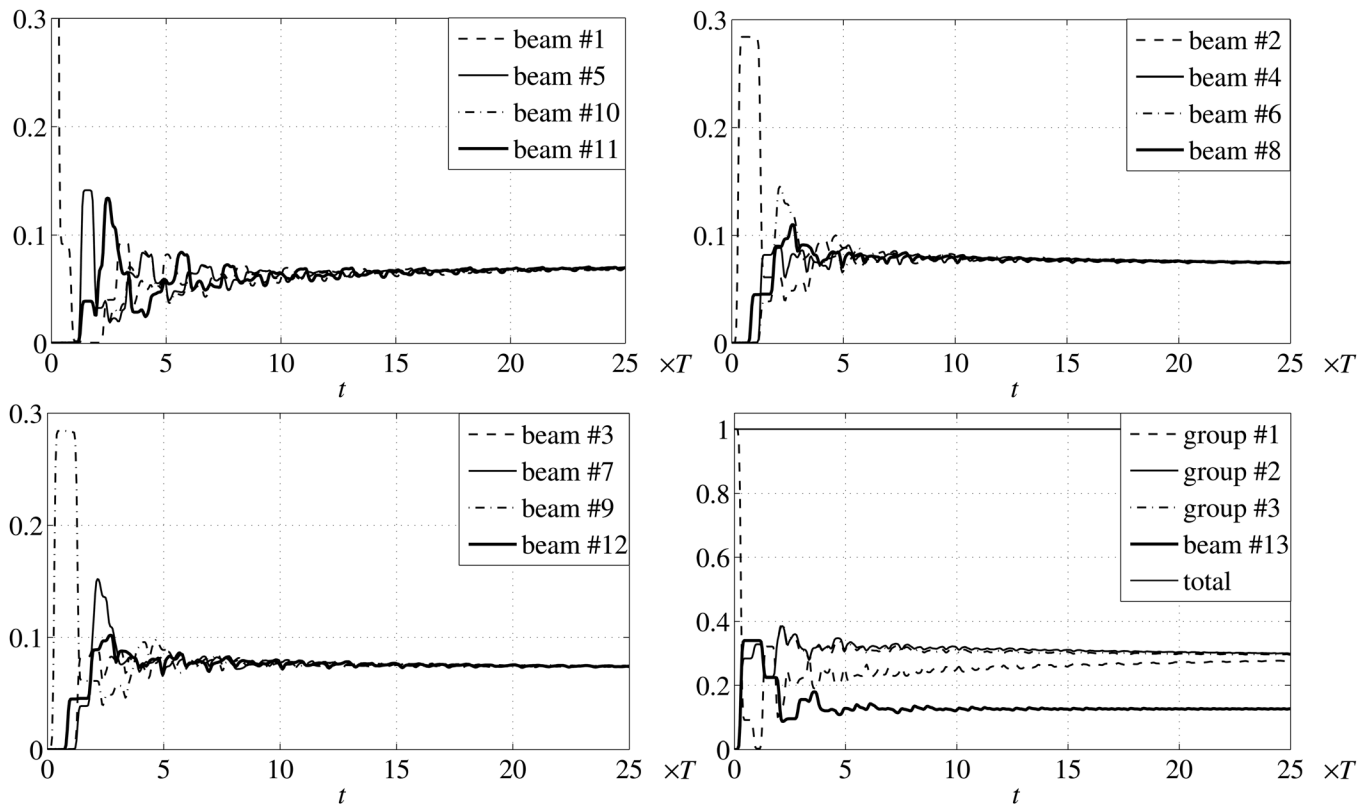


FIG. 9. Evolution of the total vibrational energy $\mathcal{E}^p(t)$, $1 \leq p \leq 13$ in the truss for an initial compressional wave. The top-left plot is for the beams parallel to beam 1 (beam group 1), the top-right plot is for the vertical beams (beam group 2), and the bottom-left plot is for those horizontal beams orthogonal to beam 1 (beam group 3). The bottom-right plot displays the evolution of the sum of the total vibrational energy for those groups, and the evolution of the total vibrational energy for beam 13.

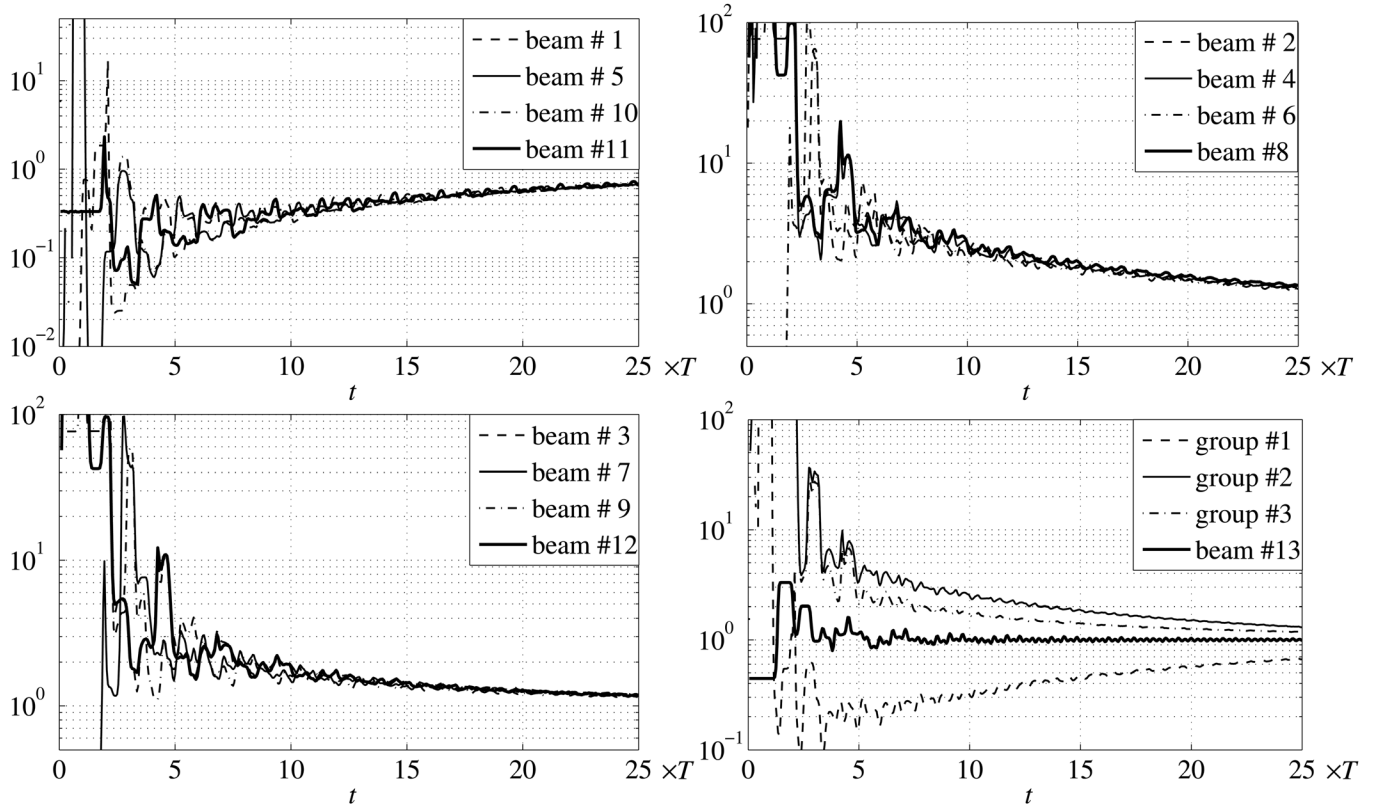


FIG. 10. Evolution of the ratio $c_T v_T^p(t) / 2 c_p v_p^p(t)$, $1 \leq p \leq 13$, in the truss for an initial compressional wave. The top-left plot is for the beams parallel to beam 1 (beam group 1), the top-right plot is for those vertical beams (beam group 2), and the bottom-left plot is for those horizontal beams orthogonal to beam 1 (beam group 3). The bottom-right plot displays the evolution of the averaged ratio in each group, and the evolution of this ratio for beam 13.

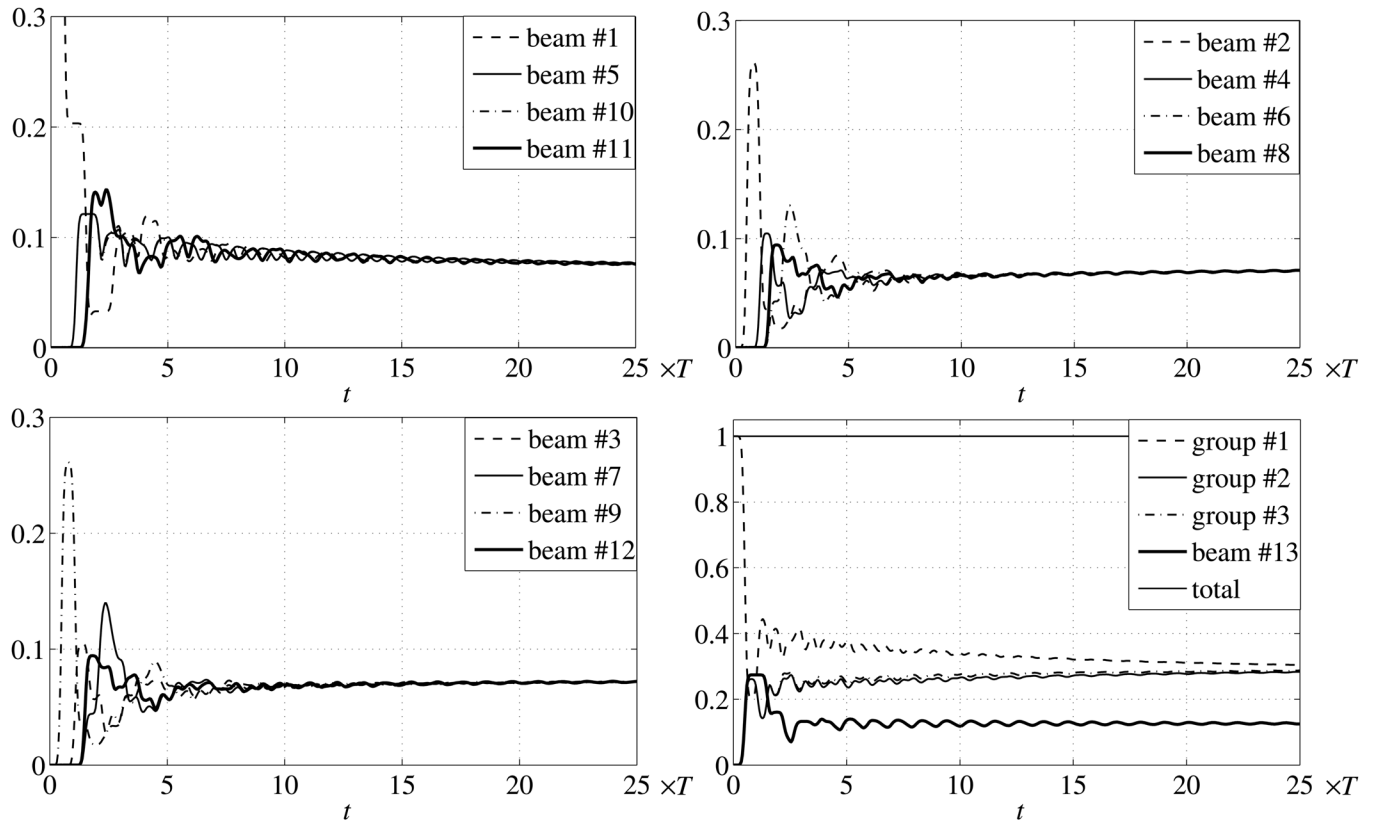


FIG. 11. Evolution of the total vibrational energy $\mathcal{E}^p(t)$, $1 \leq p \leq 13$, in the truss for an initial torsional wave. The top-left plot is for the beams parallel to beam 1 (beam group 1), the top-right plot is for the vertical beams (beam group 2), and the bottom-left plot is for those horizontal beams orthogonal to beam 1 (beam group 3). The bottom-right plot displays the evolution of the sum of the total vibrational energy for those groups, and the evolution of the total vibrational energy for beam 13.

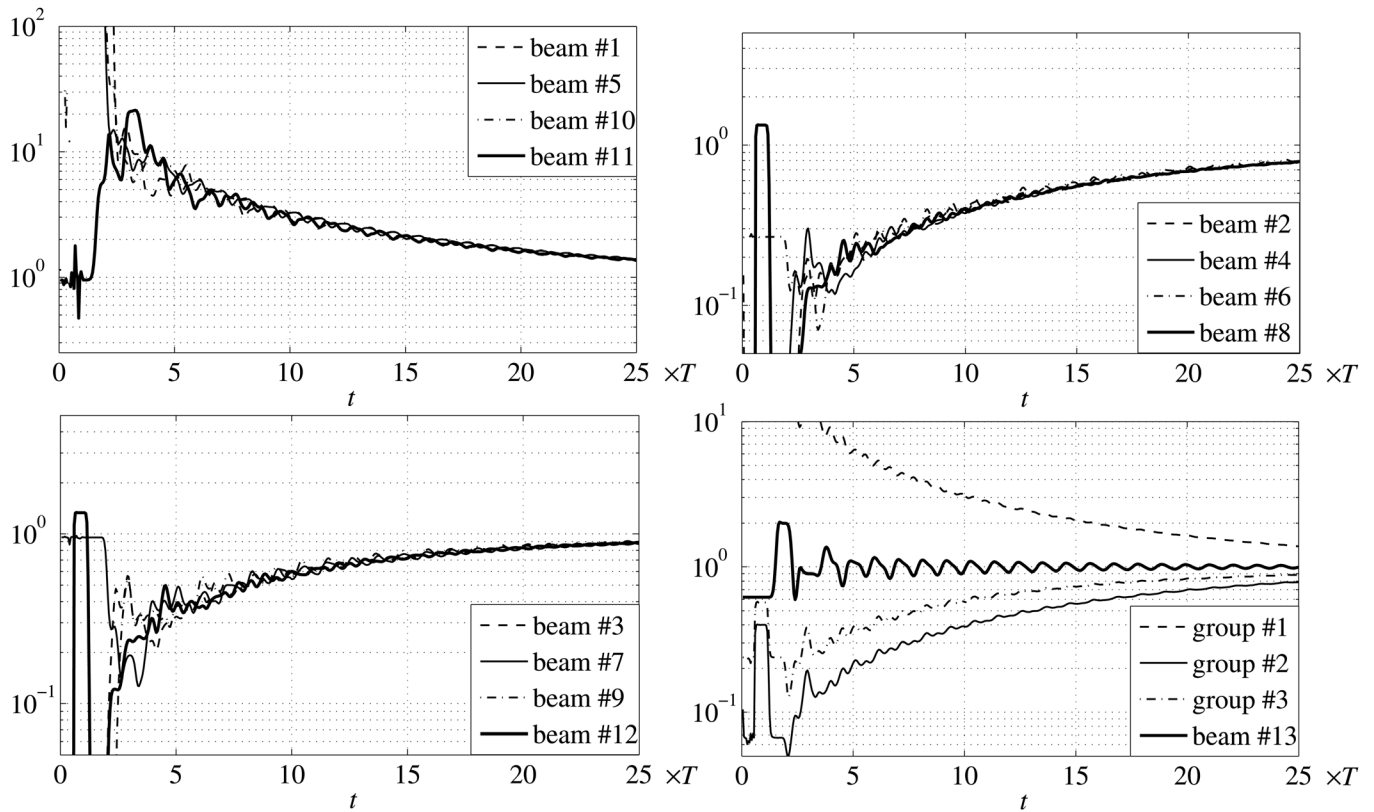


FIG. 12. Evolution of the ratio $2c_T v_T^p(t) / C P v_p^p(t)$, $1 \leq p \leq 13$ in the truss for an initial torsional wave. The top-left plot is for the beams parallel to beam 1 (beam group 1), the top-right plot is for the vertical beams (beam group 2), and the bottom-left plot is for those horizontal beams orthogonal to beam 1 (beam group 3). The bottom-right plot displays the evolution of the averaged ratio in each group, and the evolution of this ratio for beam 13.

mode is the torsional mode $\alpha = T_3$ and the longitudinal modes are the pure bending modes $\alpha = P_2$ or P_3). It is observed that the above energy ratios apparently converge to the expected diffusion limit. In addition, the second and third beam groups have the same evolution of the total vibrational energy (with a delay), similar to the previous load case. It should be noted that the high variations observed at the beginning of the simulation occur because some beams are not loaded yet or loaded with compressional or transverse waves solely.

VI. CONCLUSION

A transient transport model for the propagation of the high-frequency vibrational energy density in three-dimensional beam trusses has been derived. It accounts for the power flow reflection and transmission phenomena at the junctions in this range. The proposed analytical model is usable in a RKDG finite element scheme. Owing to its low numerical dispersion and dissipation errors, the RKDG method is suitable for long time simulations. Numerical results obtained for an example of a beam truss illustrate the transient regime and the onset of a diffusive regime at late times. The latter is prompted by the multiple reflections and transmissions of the energy modes at the beam junctions, with possible mode conversions. This scattering process is different from the multiple scattering of waves on random heterogeneities or inclusions considered elsewhere,^{30,34,36,53,54} though it gives rise to the same qualitative behavior at late

times. As observed elsewhere,⁴⁹ the outbreak of diffusion tends to be only accelerated by the presence of random heterogeneities. Quantitatively, randomness would, however, contribute to mix all wave modes, whereas the considered structure does not because it has some symmetry, as explained in the foregoing section. Diffusion is characterized by an equilibration of the total energy in each subsystem (the beam components of the truss) and for each propagation mode. This situation is comparable to the assumption of modal equipartition invoked in the SEA of structural-acoustic systems. However, a rigorous mathematical model of the diffusive behavior induced by multiple scattering at the junctions and interfaces is lacking at present. Other improvements of the proposed model could be the enrichment of the kinematic assumptions in order to describe warped cross sections, or the addition of prestressing forces. An analysis with more complex junctions should also be considered in future works, including for example elastic constraints or dissipation phenomena. These issues are the subject of ongoing researches.

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